

INTEROPERABILITY MATURITY LAB (IML)

Founding Manuscript

A Scientific Framework for Trusted Health Information Ecosystems

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Author	Dr Renato Barrios
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Health is our purpose. Trustworthy information is our foundation. Interoperability is our path.

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How to Read this Manuscript

Established evidence. Statements intended to be supported by peer-reviewed literature, international recommendations, official documentation, standards or validated public evidence.

IML proposal. Concepts, architectures and methodological choices proposed by IML for discussion, testing and refinement.

Research hypothesis. Ideas that appear plausible and useful but require empirical validation.

Reference implementation. Software structures or modules that exist as implementable work but are not automatically clinically validated.

Future work. Sections, indicators, country adaptations, connectors or studies that remain to be completed or externally validated.

Opening Statement

Health is humanity's common objective.

Progress in health information should not be measured by the sophistication of technology alone, but by its capacity to help people make better decisions, reduce suffering and improve health together.

Every day, clinical observations, laboratory analyses, medical images, prescriptions, epidemiological reports, administrative records and scientific discoveries contribute to our understanding of health. Yet information alone does not protect health.

Information protects health only when it becomes trusted knowledge, available to the right person, at the right time, under the right conditions.

This transformation cannot be achieved by technology alone. It requires scientific knowledge, institutions, governance, trust, professional responsibility and collaboration.

The Interoperability Maturity Lab (IML) was created to support this ambition. It proposes a scientific framework for understanding, assessing and progressively improving Health Information Ecosystems, together with an open reference implementation pathway in which the underlying principles can be tested.

This edition records a transition from concept toward testable implementation. PostgreSQL clinical structures, terminology services, country-pack logic, local structured clinical extraction and traceable evidence-retrieval architecture now provide concrete objects for scientific review. These developments do not constitute a claim of validated clinical deployment.

Preface

Healthcare has entered an unprecedented digital era. Hospitals, primary care practices, laboratories, pharmacies, insurers, public-health authorities and research institutions generate and exchange vast quantities of health information every day. Yet despite remarkable technological progress, healthcare professionals, patients and decision-makers continue to face a familiar challenge: information often remains fragmented when it is most needed.

The consequences are tangible. Clinical decisions may be made with incomplete information. Diagnostic procedures may be unnecessarily repeated. Prevention opportunities may be missed. Public-health surveillance may identify resistant microorganisms without fully understanding the clinical situations in which they occur. Even in highly digitalised systems, continuity of information frequently remains more difficult to achieve than continuity of care itself.

This observation is not a criticism of existing initiatives. WHO, HL7 International, SNOMED International, openEHR, LOINC, IHE, national health authorities and many other organisations provide indispensable foundations. IML is intended to be complementary rather than duplicative.

IML is also complementary to the Global Digital Health Monitor and other digital health maturity initiatives. High-level national indicators can describe strategy, infrastructure and capacity. IML asks a narrower and more clinically oriented question: does that capacity preserve meaning, context, trust, correction and usefulness across the Health Information Ecosystem?

Health is the objective. Information is the foundation. Interoperability is the path. Digital technologies have no intrinsic health value unless they contribute to prevention, safer diagnosis, more appropriate treatment, rehabilitation, research, public health or continuity.

The first operational demonstrator remains antimicrobial resistance and multidrug-resistant bacteria because it exposes the difference between data availability and clinical meaning. A resistant isolate is not automatically a symptomatic infection. Linking microbiology to symptoms, diagnosis, treatment and outcomes requires real interoperability.

Artificial intelligence is treated as a potentially powerful instrument and as an ecosystem actor. It may help organise information, retrieve evidence, identify omissions or support analysis. It must not conceal uncertainty, provenance or responsibility. This manuscript benefited from editorial and analytical assistance from OpenAI ChatGPT; authorship and scientific responsibility remain with the author.

New in this edition, the AI architecture is described in two layers. AI-EVIDENCE-01 is the implemented retrieval and provenance scaffold for scientific sources. AI-EVIDENCE-02A is proposed as a patient-independent clinical knowledge layer that preserves source provenance, studied populations, evidence certainty, jurisdictional authority and applicability before information is presented in a consultation. It includes a medication decision-support direction for fixed-dose combinations, interactions and cumulative risk. The second layer is explicitly presented as a proposal and migration draft, not as a validated clinical service.

The IML Declaration

Health is one of humanity's greatest shared responsibilities. Every healthcare professional, patient, researcher, institution and policymaker contributes, directly or indirectly, to its protection.

IML is an independent scientific initiative dedicated to understanding, assessing and progressively improving Health Information Ecosystems. It does not promote a particular software platform, technical standard, commercial product or national model. It seeks an open scientific framework capable of supporting continuous improvement through evidence, collaboration and practical implementation.

Interoperability is not an objective in itself. Its purpose is to enable trustworthy information to circulate safely, meaningfully and responsibly wherever it is required to support prevention, diagnosis, treatment, rehabilitation, research and public health.

Technical interoperability alone is insufficient. Sustainable progress also depends on governance, institutional engagement, semantic consistency, legal certainty, professional collaboration, public trust, security, correction and learning.

Health systems differ considerably. Economic resources, institutional structures, financing models, legal frameworks and digital maturity vary widely. IML therefore rejects the idea that maturity is simply conformity with a single technological architecture.

Public insurers, national health services, complementary insurers and private payers are integral actors because their longitudinal information may support continuity, prevention and evaluation when governance is appropriate.

Artificial intelligence may assist professionals, researchers and institutions, but scientific judgement, ethical responsibility and clinical decision-making remain human responsibilities. AI should be auditable, contestable and proportionate to its intended purpose.

No institution or country can achieve interoperability alone. Progress depends on openness, cooperation, correction and continuous learning. Every ecosystem can improve, and every documented failure can become an opportunity to learn.

The IML Commitment

- Put health before technology.
- Promote trustworthy, contextualised, secure and correctable health information.
- Support continuity of health information across care, public health, research and financing.
- Encourage open standards and open-source reference implementations when they improve transparency, autonomy and scientific review.
- Evaluate ecosystems and relationships rather than isolated applications.
- Distinguish documented implementation from aspiration and documentary confidence from performance.
- Treat feedback, correction, documentary reliability and institutional answerability as observable dimensions of learning.
- Keep artificial intelligence under human responsibility and preserve provenance from clinical question to evidence and interpretation.
- Support progressive deployment in both high-resource and constrained settings.
- Prefer continuous improvement and evidence profiles over simplistic country rankings.

PART 0 - Why This White Paper Exists

The purpose of this manuscript is not to introduce another digital health initiative. It is to understand how health information can better protect health.

0.1 Why another white paper?

Healthcare is experiencing one of the most profound transformations in its history. Electronic health records, laboratory systems, imaging, genomics, national digital services and artificial intelligence have become ordinary parts of healthcare. These developments rest on major international work by WHO, HL7, SNOMED International, openEHR, IHE, LOINC, the OECD, the European Commission and national agencies.

Despite this progress, health information often remains fragmented when it is most needed. Clinicians may make decisions with incomplete histories; patients may repeat investigations; professionals may spend time searching instead of interpreting; researchers may lack structured context; and public-health bodies may receive technically correct data that are clinically ambiguous.

The problem is therefore no longer simply digitalisation. It is transformation: how can data become trusted knowledge that contributes to health? This question is the starting point of IML.

0.2 A changing paradigm

Digital health first focused on computerising care. A second phase focused on connecting systems. A third phase is now emerging in which the decisive test is whether information improves decisions.

This shift changes the unit of analysis. An application can be technically successful while the surrounding ecosystem still fails to preserve meaning. Conversely, relatively simple technologies may produce high value when they are well governed, clinically aligned and capable of correction.

IML therefore treats technology as an instrument. Health remains the objective, and interoperability is valuable only insofar as it helps transform trustworthy information into useful knowledge.

0.3 Why Health Information Ecosystems?

Healthcare is frequently described through individual information systems: hospital systems, electronic records, laboratory systems, pharmacy platforms and registries. Each has a legitimate purpose, yet health rarely follows organisational boundaries.

Patients move, professionals collaborate, biological samples circulate, scientific knowledge evolves and financing organisations observe longitudinal trajectories. The relevant object is therefore the network of relationships among actors, data, standards, institutions and responsibilities.

IML uses the term Health Information Ecosystem to describe this wider socio-technical environment. Ecosystem maturity depends less on possession of a particular product than on the capacity of the whole network to preserve trustworthy information.

0.4 Beyond technical interoperability

An additional difficulty is that the different dimensions of interoperability can mature at different speeds. A country may standardise identifiers before it has consistent terminology services, or build national APIs before local software vendors have the resources to implement them reliably. The presence of an exchange specification is therefore only one point in a chain that includes procurement, certification, local configuration, professional workflow and maintenance. IML attempts to keep these stages visible so that a policy success is not mistaken automatically for operational success.

This broader view also protects against a common conceptual shortcut: attributing every interoperability problem to the absence of a technical standard. Sometimes the standard exists and the bottleneck lies in governance, incentives, role definition, licensing, security, documentation or institutional response. The framework should therefore help investigators locate the real level of failure before recommending another layer of technology.

Technical, semantic and organisational interoperability are essential, but practical experience shows that they do not automatically guarantee meaningful collaboration. A technically valid message can still arrive too late, lack context, be poorly trusted or remain uncorrected.

Successful interoperability also depends on institutional commitment, governance, legislation, education, professional trust, cybersecurity and sustained cooperation. These conditions shape whether technical capacity becomes routine clinical usefulness.

IML therefore extends assessment beyond connection itself. It asks whether the ecosystem can use, question, correct and learn from the information it exchanges.

0.5 A practical scientific framework

IML does not seek to replace standards. It seeks to organise evidence about their practical use around health-oriented questions.

Every assessment should answer three questions: where are we today, what should improve next, and how will progress be demonstrated? A framework that produces a score without an improvement pathway is of limited value.

This practical orientation is why IML combines maturity assessment, country evidence profiles, a focused AMR demonstrator and an Open Clinical Workspace reference implementation.

0.6 The role of public and private payers

Health financing organisations possess information about eligibility, utilisation, reimbursement, chronic-care pathways and longitudinal service use. In many systems these data remain separated from clinical and public-health information.

IML treats Payer Interoperability as a cross-cutting dimension because payers differ by country. A national health service, statutory social insurance, mutual insurer and private insurance company cannot be treated as identical institutions, yet each can influence information continuity.

The objective is not unrestricted sharing. It is proportionate, governed exchange when payer-held information can contribute to continuity, prevention, evaluation or administrative coherence.

0.7 Why antimicrobial resistance?

Every scientific framework requires practical validation. IML proposes antimicrobial resistance, particularly multidrug-resistant *Escherichia coli* in urinary tract infection, as its first demonstrator.

AMR exposes the limitations of fragmented information. Microbiology alone cannot distinguish colonisation from infection. An antibiogram does not establish clinical significance. Meaningful interpretation requires symptoms, specimen context, diagnosis, treatment and outcome.

The use case therefore tests technical exchange, semantic consistency, clinical context preservation, public-health learning and the possibility of linking evidence retrieval to a concrete clinical question.

0.8 A framework for every country

Countries differ in resources, institutions, financing models, legal environments and digital infrastructure. IML does not assume that one implementation path fits all.

A low-resource setting may reasonably prioritise reliable patient identification, simple structured laboratory exchange and offline capability before advanced analytics. A highly digitalised system may need to focus on fragmentation, proprietary dependencies or institutional follow-through.

Maturity therefore means progressive capacity to learn and improve, not conformity with a single ideal architecture.

0.9 Complementarity with the Global Digital Health Monitor and digital health maturity initiatives

Country-level maturity instruments remain extremely useful because they provide a common vocabulary for leadership, strategy, infrastructure and services. IML adds value when it descends from national capacity to the information relationship itself. A national patient portal may exist, for example, while a laboratory result still lacks the context required for a safe clinical interpretation. The two observations are not contradictory; they describe different levels of the same ecosystem.

For this reason, IML should preserve the validation status and date of external sources. Digital-health dashboards evolve, governments update national strategies and institutions may correct published documentation. A country profile should therefore be reproducible as of an assessment date. Later evidence should create a new version rather than retroactively changing what reviewers could have known at the time.

The Global Digital Health Monitor and related maturity tools provide valuable high-level views of strategy, governance, services, infrastructure, standards and workforce. IML should use such information as context rather than duplicate it.

National dashboards often rely on official reporting cycles whose validation status can vary over time. IML therefore records the source, date, authority and limitations of evidence rather than assuming that a published national statement proves routine implementation.

The complementary IML question is whether digital capacity preserves meaning, context, trust and clinical usefulness in concrete pathways. Where evidence is incomplete, IML records uncertainty and invites review rather than silently filling gaps.

0.10 Towards a shared future

Pandemics, antimicrobial resistance, ageing, chronic disease, migration and artificial intelligence demonstrate that health challenges cross organisational and national boundaries.

The future therefore depends not only on better technology but on stronger cooperation, reproducible evidence and the capacity to correct systems when they fail.

IML is proposed as a contribution to this collective effort. It does not claim to possess all the answers. It seeks a framework in which better questions can be asked, better evidence generated and better decisions supported.

Key Messages

- Digitalisation is not the same as clinically useful interoperability.
- Health Information Ecosystems are the appropriate unit of analysis.
- IML complements rather than replaces established standards and national maturity monitoring.
- AMR/BMR provides a demanding first demonstrator of context preservation.
- Country diversity requires progressive improvement rather than one universal implementation model.

PART I - Health Before Technology

Technology changes rapidly. Health remains humanity's enduring objective.

1.1 Beginning with Health

Most digital-health publications begin with software, platforms or standards. IML deliberately begins elsewhere: the purpose of healthcare is not to build digital systems but to protect and improve health.

Electronic records, laboratory systems, AI and interoperability standards acquire value only when they contribute to prevention, safer diagnosis, effective treatment, stronger public health, research or continuity. This health-first orientation governs the framework.

1.2 The WHO Definition of Health

The WHO Constitution defines health in physical, mental and social terms rather than as the absence of disease alone. This broader conception matters for information architecture because no single institution or dataset can represent the whole of health.

Clinical records must therefore coexist with information about function, prevention, social circumstances, public health and longitudinal experience when such information is relevant and lawfully used.

1.3 Health as a Dynamic Continuum

Health is not a binary state. People move along a dynamic continuum influenced by ageing, behaviour, genetics, environment, healthcare, social conditions and unexpected events.

Information systems should therefore support trajectories, not merely episodes. Longitudinal patterns often matter before a formal diagnosis exists, and continuity of information can create opportunities for earlier intervention.

1.4 Prevention and Early Detection

Preventive usefulness also depends on avoiding excessive surveillance and false alarms. Combining more data can improve detection, but it can also produce noise, overdiagnosis or unnecessary intervention when thresholds and clinical context are poorly defined. The health-first principle therefore requires proportionality: a signal should lead to an appropriate review process rather than automatic escalation simply because it was generated digitally.

Prevention illustrates why information quality and continuity are inseparable. A single abnormal value may be uninformative, while a trajectory across several years can reveal a meaningful change. Conversely, a longitudinal record can be misleading if laboratory methods, units, reference ranges or patient identity are inconsistent. Mature prevention therefore requires both continuity and semantic discipline.

Many important conditions develop progressively. Before diagnosis, weak signals may already exist in laboratory values, repeated consultations, medication patterns, admissions or patient-reported symptoms.

These signals often remain dispersed across organisations. Prevention can fail because the information exists but cannot be assembled with sufficient context. Interoperability should therefore support earlier, proportionate recognition without turning every data point into an alarm.

1.5 Information as a Determinant of Health

Traditional determinants of health include biology, environment, socioeconomic conditions and healthcare access. IML proposes trustworthy information as an enabling determinant because it changes the quality and timing of decisions.

Information does not improve health by itself. Its effect is mediated through decisions, professional action, patient understanding and public-health response. The relevant property is therefore trustworthy usefulness, not volume.

1.6 Continuity of Care and Continuity of Information

Continuity has several dimensions. Informational continuity concerns what clinicians know about prior events. Management continuity concerns coherence across successive decisions. Relational continuity concerns the therapeutic relationship itself. IML focuses particularly on informational continuity while recognising that the three dimensions interact. Better information cannot replace a trusted professional relationship, but missing information can undermine it.

The architecture should therefore support retrieval without assuming that every datum must be permanently centralised. Federated access, local storage, controlled import and patient-mediated access can all contribute, depending on context. The scientific question is whether the necessary information can be obtained with enough reliability, timeliness, provenance and legal clarity to support the next decision.

Continuity of care depends on professionals having sufficient knowledge of previous diagnoses, treatments, preferences and outcomes. When patients cross organisational boundaries, that continuity is threatened by fragmented information.

IML therefore introduces Continuity of Health Information: the capacity of relevant information to accompany the health journey across time, settings and authorised actors without losing meaning, integrity or clinical relevance.

Continuity does not require universal centralisation. It requires reliable retrieval, provenance, governance and the ability to understand what the information meant in its original context.

1.7 From Data to Knowledge

Healthcare produces extraordinary quantities of data. Data become useful only after interpretation. Information becomes knowledge when it can be connected to context, evidence and purpose.

Interoperability should therefore be evaluated by whether it supports trustworthy knowledge, not merely whether two systems can exchange a payload.

AI may assist this transformation, but it cannot repair missing provenance simply by producing a plausible narrative.

1.8 Health Information Ecosystems

Patients, clinicians, hospitals, laboratories, pharmacies, public-health authorities, universities, payers, industry and AI systems all contribute to the information environment.

The maturity of this environment depends on the quality of relationships between actors: governance, standards, trust, accountability, context, correction and learning.

1.9 Why Technology Alone Is Not Enough

Technological progress remains essential, but many failed or disappointing digital-health initiatives demonstrate that installation is not transformation.

Successful adoption requires governance, professional alignment, training, legal certainty, incentives, maintainability, cybersecurity and the ability to learn from incidents.

IML therefore treats interoperability as a socio-technical capability.

1.10 Towards Health Information Ecosystems

Health remains the objective, information provides the foundation, and interoperability is one of the practical conditions required to transform fragmented observations into trustworthy knowledge.

The following parts therefore move from individual systems toward the broader ecosystem in which information is created, exchanged, interpreted, corrected and reused.

1.11 Health Information and Health Equity

Health information is not distributed equally. Some people receive care in highly connected environments, while others move through fragmented services in which clinicians repeatedly reconstruct history.

Fragmentation can amplify existing inequalities, especially for people with chronic disease, disability, multiple comorbidities or unstable access to care. Information equity therefore matters alongside service access.

IML does not assume that more data automatically create equity. It asks whether the information required for safe and fair care is accessible under appropriate governance.

1.12 Information Across the Entire Health Journey

Health information begins before disease appears and continues through prevention, screening, acute care, rehabilitation, chronic follow-up, palliative care, public health, research and financing.

IML describes this as the Health Information Lifecycle. Information is generated, validated, interpreted, shared, corrected, reused, preserved and eventually archived or deleted according to purpose and law.

AI increasingly participates in several stages, which makes provenance and lifecycle governance even more important.

1.13 Health Information and Public Trust

Public trust is also affected by the way institutions respond when something goes wrong. Security incidents, erroneous records, inaccessible services and contradictory official documentation are inevitable in complex systems. Trust is strengthened when institutions acknowledge problems, provide a correction pathway, explain what changed and demonstrate learning. Silence or opaque remediation can be more damaging than the original technical failure.

This observation connects the health-first philosophy to the later Learning domain. Trust is not a static reputation score. It is produced continuously through the experience of using, questioning and correcting the ecosystem. A mature system should make its own fallibility manageable.

No Health Information Ecosystem can function without trust. Patients must trust that information will be handled responsibly; professionals must trust what they receive; researchers must trust data quality; institutions must trust each other's governance.

Trust is not equivalent to confidence in a brand. It grows from transparency, accountability, provenance, security, correction and the ability to explain limitations.

A mature ecosystem should make errors correctable rather than pretending they do not occur.

1.14 Medical Error, Patient Safety and Information Failure

Patient safety provides a direct test of the health-first principle. WHO identifies avoidable adverse events, errors and risks associated with healthcare as continuing global patient-safety challenges. IML therefore treats medical error not only as an individual professional event, but also as a possible ecosystem failure in which information, workflow, software, institutions and scientific knowledge interact.

An error may emerge from incomplete or fragmented information, time pressure, therapeutic complexity, polypharmacy, ambiguous responsibility, poorly calibrated decision support, inaccessible source material or limits in the evidence available for the person being treated. This framing does not remove individual professional responsibility and does not attribute a particular incident to a system cause without evidence. It broadens the field of investigation so that preventable information and organisational contributors can be examined.

IML proposes a comparative research workstream on medical error, patient redress and learning across health systems. The study should examine how harm is detected, disclosed, reported, reviewed and corrected; how patients can obtain explanation, compensation or other redress; how legal and administrative pathways differ;

and whether incident information produces durable changes in practice, software or governance. The consequences for the affected person, including time, uncertainty and burden of navigating a remedy pathway, are part of the research question.

No comparative conclusion is drawn in this edition. Published legal routes, institutional descriptions and compensation mechanisms should not be treated automatically as evidence that a remedy is timely, accessible or effective in practice. The workstream should triangulate official information, independent audit, published outcomes and, where appropriate, patient or professional experience before drawing conclusions.

For IML, the central patient-safety question is whether an ecosystem can make clinically relevant information available before a decision, identify important missing context, make errors visible and correctable, and transform incidents into learning without displacing professional judgement.

1.15 Towards a New Vision of Interoperability

The traditional question asks whether systems can exchange data. IML asks whether the ecosystem can transform trustworthy information into knowledge that better protects health.

This shifts the unit of analysis from application to ecosystem, the objective from technology to health, and the measure of success from message volume to decision usefulness.

The question that guides the manuscript is therefore deliberately simple: do our information arrangements help people remain healthier?

Key Messages

- Health precedes technology.
- Prevention depends on meaningful continuity of information.
- Trustworthy information is an enabling determinant of health.
- Health equity can be harmed by fragmentation.
- Interoperability should be judged by preservation of meaning and contribution to decisions.

- Medical error is also an information, workflow and learning problem under professional responsibility.

PART II - Health Information Ecosystems

2.1 From Information Systems to Health Information Ecosystems

Healthcare has digitised rapidly, yet patients continue to move across the organisational boundaries around which information systems were built. A general practitioner, hospital, laboratory, pharmacy, rehabilitation service and payer may each hold a legitimate fragment of the same health journey.

The paradox is that healthcare can become more digital while clinical knowledge remains fragmented. The solution is not necessarily one universal system, but an ecosystem capable of preserving meaning across independent systems.

IML therefore uses the ecosystem as its fundamental unit of analysis.

2.2 Defining the Health Information Ecosystem

The ecosystem concept deliberately avoids assuming a single centre. Some countries organise health information through strong national services; others rely on regional or institutional infrastructures; many combine both. IML can be applied to each configuration because it evaluates the relationships by which information becomes usable rather than prescribing a central topology.

The ecosystem also contains actors that do not directly provide care. Standards organisations, software maintainers, cybersecurity teams, cloud or database providers, auditors and courts of audit may all influence whether information remains available and trustworthy. Their role becomes especially visible when systems fail, require correction or must be migrated.

A Health Information Ecosystem is the dynamic network of individuals, organisations, technologies, governance mechanisms and information flows that collectively generate, validate, exchange, interpret and use trustworthy health information in order to protect and improve health.

The word dynamic is important. Scientific knowledge, regulation, clinical practice, software and institutional responsibilities change. An ecosystem must therefore be capable of adaptation and correction.

The definition also makes purpose explicit. The ecosystem exists for health, not for technological optimisation in isolation.

2.3 The Actors of the Ecosystem

Patients produce information through symptoms, experiences, preferences, outcomes and increasingly personal devices. Professionals transform observations into clinical reasoning. Laboratories contribute biological evidence. Hospitals coordinate complex multidisciplinary care. Public-health authorities interpret populations. Universities transform information into scientific knowledge.

Payers provide longitudinal administrative and utilisation perspectives; industry contributes medicines, devices and digital systems; standards organisations maintain shared vocabularies and exchange specifications; regulators define legal and safety conditions.

Artificial intelligence is a new actor whose outputs can influence what humans notice or decide. Its role must therefore be visible, governed and attributable.

2.4 Relationships Rather than Organisations

Patients experience transitions, not organisational charts. Every transition creates an information relationship: hospital to primary care, laboratory to clinician, clinician to pharmacy, insurer to provider, surveillance system to public-health action.

The maturity of the ecosystem depends on these relationships. A technically excellent hospital system can coexist with poor ecosystem maturity if discharge information is delayed, laboratory context is lost or corrections do not propagate.

IML therefore evaluates relationships through continuity, trust, semantics, response, security and shared learning.

2.5 Information Flows: From Data Production to Shared Knowledge

Health information flows are not simple one-way transfers. Information is produced, transformed, interpreted, queried, corrected and reused by different actors for different purposes.

A laboratory result can support an immediate treatment decision, later quality review, surveillance, research or evidence generation. Each reuse needs context, governance and provenance.

Shared knowledge emerges only when information can be understood beyond the system that originally created it.

2.5.1 The Health Information Lifecycle

Lifecycle thinking also clarifies why provenance cannot be added only at the moment of research reuse. By the time information reaches a secondary analytics environment, the original source, correction history or unit transformation may already have been lost. Provenance should therefore be captured as close as possible to the original event and preserved through later transformations.

The same principle applies to AI. A model-generated summary is a new information object with its own authoring process, model version, inputs and review status. Treating it as indistinguishable from source clinical data would collapse two different stages of the lifecycle and make later audit much more difficult.

The lifecycle begins when information is generated and continues through validation, storage, exchange, interpretation, correction, reuse and preservation. Deletion or archival may also be part of the lifecycle.

Every stage can introduce error. Patient matching can fail, coding can drift, interfaces can truncate context, or a later correction may remain confined to the originating system.

Maturity therefore requires lifecycle controls rather than a single point-in-time interoperability test.

2.5.2 Information Has Context

An observation acquires meaning through the circumstances in which it was produced. A positive urine culture means something different in a symptomatic patient, an asymptomatic patient, a contaminated specimen or a screening context.

Context includes who, when, why, where, under which method, with which uncertainty and with what subsequent interpretation or outcome.

Clinical context preservation is therefore a central IML concept.

2.5.3 Information Quality

Database integrity is one part of information quality, but not the whole. A database can enforce a valid date, code and foreign key while still storing a clinically misleading interpretation. Conversely, unstructured narrative may contain essential nuance even when it is difficult to analyse computationally. IML therefore distinguishes structural quality from clinical fitness for purpose.

Quality also changes with reuse. A field created for billing may be accurate for reimbursement and still be too coarse for research. A laboratory code may be appropriate for local operations and require mapping before cross-border comparison. Reuse should therefore carry the original purpose, source and transformation history whenever those characteristics influence interpretation.

Information quality includes accuracy, completeness, timeliness, consistency, provenance, interpretability and fitness for purpose. No single quality metric is sufficient for every use.

A field that is adequate for reimbursement may be insufficient for clinical decision support. A laboratory code may be technically valid but clinically misleading if the specimen or reference method is missing.

Quality should be evaluated in relation to intended use and the consequences of error.

2.5.4 From Information to Shared Knowledge

Shared knowledge is not simply a shared database. It is the capacity for multiple actors to interpret information coherently and act appropriately.

Terminology, contextual metadata, provenance, professional judgement and evidence are all required. Shared knowledge remains contestable and correctable as new information appears.

IML therefore links semantic interoperability to learning rather than treating codes as final truth.

2.5.5 Knowledge Continuity

Knowledge Continuity describes the capacity of successive clinical, public-health and research activities to build on previous validated knowledge rather than repeatedly restarting from fragmented information.

This requires durable provenance, versioning and correction. A later clinician should be able to know what was believed, on what basis, and whether that interpretation changed.

AI systems should inherit this history rather than generate a decontextualised summary.

2.6 Governance of Health Information Ecosystems

Governance must also manage change. Standards are revised, national services are replaced, software reaches end of support and new legal requirements appear. Mature governance should therefore include version policies, migration planning, communication with affected users and explicit responsibility for backward compatibility or deprecation.

A governance body should also be able to distinguish strategic authority from technical operation. The institution that selects a national standard may not be the institution that implements an interface, certifies software or manages incidents. Making these responsibilities explicit prevents problems from being transferred indefinitely between organisations.

Governance defines responsibility, authority, accountability, priorities and acceptable use. It should clarify who can create, access, correct, link, analyse and publish information.

Effective governance is not only a policy document. It includes implementation mechanisms, decision records, escalation pathways, incident management and transparent updating of rules.

Open-source software also requires governance: code availability does not replace maintenance, clinical safety review or accountable stewardship.

2.7 Trust as Infrastructure

Trust functions as an invisible infrastructure. Systems that cannot explain provenance or correction may be technically connected but operationally distrusted.

Trust must be earned through security, accountability, predictable governance, professional competence and transparency about limitations.

IML therefore treats trust both as a prerequisite and an outcome of maturity.

2.8 Institutional Engagement

Interoperability depends on organisations being willing and able to participate in dialogue, governance, pilots and corrective action. Contact points and procedures matter because they determine whether collaboration is even possible.

Institutional Engagement therefore evaluates willingness and practical participation. It is broader than responsiveness to a single inquiry and should not be confused with the Learning-domain criterion LRN-5.

An organisation may have excellent formal processes but still fail to provide substantive follow-through. The framework keeps these phenomena distinct.

2.9 Institutional Readiness

Institutional Readiness refers to practical preparedness to collaborate, allocate resources, align incentives, respond to implementation needs and sustain interoperable processes.

Readiness may be limited by workforce, procurement, legal uncertainty, legacy infrastructure, absent governance or competing priorities. These constraints should be described rather than moralised.

Readiness is particularly relevant when transferring a technical design from one country or institution to another.

2.10 Payer Interoperability

Payer interoperability should be evaluated with particular caution because payment systems can influence behaviour. Data that improve administrative efficiency may also create incentives that distort clinical documentation. IML should therefore examine whether coding requirements align with clinical meaning and whether professionals can correct or qualify administrative information that is later reused.

At the same time, payers may hold valuable longitudinal signals that no single provider can see, including repeated emergency use, medication reimbursement or cross-provider episodes. Under appropriate governance, these signals can support prevention and continuity. The challenge is to combine administrative reach with clinical interpretation without allowing one to substitute for the other.

Public and private payers often observe long periods of a patient's healthcare use. Their information can contribute to continuity, chronic-disease management and evaluation when clinically relevant and lawfully shared.

Yet payer data are primarily administrative and must not be mistaken automatically for clinical truth. Diagnoses used for reimbursement may not contain the context needed for clinical interpretation.

IML therefore treats payer information as one layer in the ecosystem, valuable when combined carefully with clinical and public-health sources.

2.11 Artificial Intelligence as an Ecosystem Actor

Modelling AI as an actor also means recognising dependencies outside the model itself. A service may rely on a specific cloud region, external API, proprietary embedding model, prompt template or retrieval index. Each dependency can affect availability, privacy, reproducibility or bias. The architecture should expose enough of this chain for meaningful review.

Provider independence is therefore an important design objective. IML should be able to replace one model with another while preserving clinical records, evidence sources, terminology, audit logs and human validation states. This reduces both technical lock-in and the scientific risk of making historical results impossible to reproduce when a commercial model changes.

AI can search, summarise, extract structure, prioritise information, suggest codes or identify patterns. These functions influence human attention and therefore deserve the same scrutiny as other ecosystem actors.

AI usefulness depends on the quality and provenance of its inputs. A model cannot make fragmented data trustworthy merely by expressing them fluently.

IML therefore separates durable clinical and evidence records from replaceable model services, keeps human validation explicit and treats model outputs as attributable transformations rather than invisible truth.

2.12 Research Perspectives

The ecosystem perspective generates research questions that cannot be answered by software benchmarks alone. How should continuity be measured? Which relationships most strongly affect outcomes? How can institutional responsiveness be assessed fairly? What level of evidence is sufficient to claim routine interoperability?

Implementation research, clinical informatics, public health, health services research, cybersecurity, law and social science all have roles in answering these questions.

IML should remain open to revising its concepts when empirical evidence shows that a proposed indicator is weak or a different unit of analysis is more useful.

Key Messages

- The Health Information Ecosystem is a dynamic network, not a single platform.
- Relationships and information transitions are central units of maturity.
- Context, quality, provenance and correction are required for shared knowledge.
- Institutional Engagement and Institutional Readiness are necessary socio-technical conditions.
- AI should be treated as a governed ecosystem actor whose outputs remain attributable.

PART III - Understanding Interoperability

3.1 Interoperability is not only technical

Interoperability is often reduced to the capacity of one system to send data to another. That capacity is necessary, but insufficient when information loses meaning, arrives without trust, conflicts with workflow or cannot be corrected.

IML defines mature interoperability as preservation of meaning, context, trust, accountability and usefulness across independently governed boundaries.

3.2 The Five Layers of Interoperability

A useful working model distinguishes technical, semantic, organisational, institutional and clinical/public-health interoperability. These layers overlap but highlight different failure modes.

Technical transport can succeed while semantic interpretation fails; semantic alignment can succeed while institutions do not cooperate; organisational processes can align while the information remains clinically incomplete.

Assessment should therefore consider layers together rather than assuming that progress at one level automatically transfers to the others.

3.3 Technical Interoperability

Technical conformance should be demonstrated, not inferred from marketing. Test payloads, negative cases, version negotiation, timeouts, duplicate handling and recovery after interruption all matter. A connector that works only on ideal examples may fail precisely when clinical information is most complex.

IML therefore favours reference test environments and reproducible integration fixtures. A laboratory import, for example, should be tested with missing fields, corrections, duplicated results, unusual units and invalid patient identifiers. Robustness to imperfect real-world input is part of technical maturity.

Technical interoperability concerns networks, protocols, APIs, structured files, message delivery, availability, encryption, conformance and the ability of systems to exchange information reliably.

IML is technology-neutral. A well-governed structured file may be preferable to an immature API. Mature ecosystems should nevertheless document interfaces, versions, dependencies, test environments and failure modes.

Portability and operating-system lifecycle support belong to this layer because unsupported infrastructure can break otherwise valid exchange.

3.4 Semantic Interoperability

Semantic interoperability is especially vulnerable to silent error because a syntactically valid code may look authoritative. Mapping should therefore preserve the source expression, target system, target version, mapping relationship, confidence or review status and the identity of the mapping process where relevant.

National language adaptations create an additional layer. Translations can improve usability but can also introduce false equivalence. The IML terminology workflow therefore treats translation as a review queue and keeps local or proposed designations separate from officially maintained terminology content.

Semantic interoperability requires shared concepts, units, code systems, versions and mappings. It should also preserve the distinction between a source term and a mapped target concept.

Terminology systems such as SNOMED CT, LOINC, ICD classifications and locally appropriate vocabularies may play different roles. Mappings should not create false equivalence.

The Open Clinical Workspace terminology work treats mappings as versioned, reviewable objects and supports a concise KSH97-P encounter-reason workflow alongside broader ICD coding.

3.5 Organisational Interoperability

Organisational interoperability concerns workflows, responsibilities, service boundaries and timing. A result that arrives after the decision has been made is technically interoperable but operationally weak.

Real workflow analysis should identify who creates, validates, receives, acts on and corrects information. Professional roles should be aligned with access rights and training.

Adoption problems should therefore be analysed partly as organisational design problems rather than simply as user resistance.

3.6 Institutional Interoperability

Institutional interoperability concerns governance across organisations: agreements, accountability, legal basis, funding, escalation, change management and sustained cooperation.

Institutions must also be capable of responding when documented problems are raised. Formal channels without substantive follow-through reveal a gap between procedural and operational maturity.

This is why IML keeps Institutional Engagement as a cross-cutting dimension while also adding LRN-5 Institutional responsiveness and answerability inside Learning.

3.7 Clinical and Public Health Interoperability

Clinical interoperability is demonstrated when information can safely support professional judgement. Public-health interoperability is demonstrated when individual observations can contribute to population learning without erasing clinical distinctions.

These layers are tightly connected in AMR. Surveillance of resistant isolates is valuable, but policy decisions require clarity about infection, colonisation, contamination, treatment and outcome.

Clinical and public-health interoperability therefore provide a stricter test than message exchange alone.

3.8 Clinical Context Preservation

Context preservation can be implemented through a combination of structured fields, links between observations and encounters, temporal relationships, source provenance and retained narrative. The goal is not maximal structure. It is sufficient structure to prevent the loss of clinically decisive meaning.

This principle can be tested empirically. Reviewers can ask whether a clinician receiving an imported result can answer the key questions required for action without returning to the source system. If not, the interoperability process may be technically successful but clinically incomplete.

Clinical Context Preservation is the preservation of the circumstances that give information its meaning, including symptoms, diagnosis, indication, treatment, evolution, specimen context and relevant risk factors.

Context may be structured or narrative. The goal is not to force every nuance into a code but to ensure that information reused for decisions retains enough context to remain interpretable.

AI can help detect missing context but should not invent it.

3.9 Trust, Consent and Digital Identity

Identity architecture must also support uncertainty and correction. Duplicate identities, changes of demographic information and mismatched records are normal operational risks. A mature system should provide reconciliation procedures and preserve an audit trail rather than silently merging records.

Consent mechanisms should be similarly practical. A consent policy that cannot be represented, revoked or understood in actual workflows offers limited protection. IML should therefore examine the operational relationship among identity, purpose, authorisation, professional role and audit.

Reliable identity is necessary for continuity and accountability, but identity is not the whole of interoperability. IML favours an identity-access layer that works with recognised national, local or organisational identity systems rather than creating a new personal identity number.

Purpose-limited, auditable access mechanisms can use signed tokens or verifiable credentials. A QR code may carry a temporary authorisation token or verification link, but should not expose sensitive identity or health information in clear text.

This approach is aligned with the principle of building on foundational identity systems and with privacy-preserving, context-specific access rather than permanent universal disclosure.

3.10 Limits of Interoperability

Not every dataset should be connected to every actor. Mature interoperability includes limits: purpose limitation, minimum necessary access, patient rights, professional confidentiality and the ability to refuse unsafe or unjustified reuse.

More exchange is not automatically better exchange. The quality of governance and context matters as much as technical availability.

An ecosystem should therefore be able to explain both why information is shared and why some information is not.

3.11 Transition to the IML Framework

Part III reframes interoperability as a multidimensional ecosystem capability. Technical exchange remains essential, but it becomes one component of a broader model that also includes meaning, workflow, institutional engagement, clinical context, public-health value and trust.

The following part translates this understanding into the domains through which ecosystem maturity can be documented and progressively improved.

Layer	Core question
Clinical and public health	Does information improve care, prevention, surveillance or learning?
Institutional	Are institutions ready and willing to collaborate and follow through?
Organisational	Do workflows, roles and responsibilities support action?
Semantic	Is meaning preserved across systems and contexts?
Technical	Can systems exchange information securely and reliably?

Key Messages

- Interoperability is preservation of meaning, trust, context and usefulness across boundaries.
- Technical interoperability is necessary but insufficient.
- IML uses five interacting layers: technical, semantic, organisational, institutional and clinical/public health.
- Clinical context preservation is a core safety property.
- Mature interoperability includes limits, authorisations and safeguards.

PART IV - The IML Framework

A maturity framework should not decide who is best. It should help each ecosystem understand what to improve next.

4.1 Why a maturity framework?

The preceding parts established the conceptual foundations of IML. Health is the objective. Trustworthy information is the foundation. Interoperability is the path through which information can become shared knowledge and support better decisions.

A conceptual framework alone is not sufficient. Health Information Ecosystems require instruments that make progress visible, comparable over time and actionable. A maturity framework provides such an instrument when it is designed not as a ranking device, but as a structured method for understanding strengths, weaknesses and realistic next steps.

IML therefore uses maturity assessment as a means of continuous improvement. The maturity score is not the objective. The improvement pathway is.

4.2 From ranking to continuous improvement

Many maturity models create an implicit hierarchy. Countries, organisations or programmes are compared according to scores that may be attractive for communication but insufficient for transformation. Such comparisons can be useful when they stimulate learning, but they become counterproductive when they obscure context.

IML adopts a different philosophy. Every ecosystem is historically, legally, economically and institutionally specific. A high-income country may possess sophisticated infrastructure while remaining fragmented. A lower-resource country may lack advanced systems while demonstrating strong institutional alignment and pragmatic interoperability.

Every assessment should therefore lead to three practical questions: Where are we today? What should improve next? How can progress be measured over time?

4.3 The six IML domains

The IML Framework is organised around six core domains: Governance and Standards; Technical Interoperability; Identity, Consent and Trust; Adoption and Use; Security and Resilience; Feedback, Correction and Learning.

The domains interact continuously. Governance influences technical choices. Identity and consent influence trust. Adoption determines whether standards become useful in practice. Security protects continuity. Feedback and correction allow ecosystems to learn.

Institutional Engagement, Payer Interoperability and AI Readiness remain cross-cutting dimensions because they influence more than one domain.

4.4 Domain 1 - Governance and Standards

Governance evidence should distinguish publication from stewardship. A standard that is formally adopted but has no maintained implementation guide, test environment or responsible authority may create more ambiguity than a simpler standard with strong operational support. Maturity therefore includes the capacity to maintain the rules after their initial publication.

The domain should also record how external feedback enters governance. Consultation procedures, issue trackers, change requests and published decisions can demonstrate whether standards evolve through a traceable learning process rather than through opaque administrative change.

Governance determines how decisions are made, who is responsible, how conflicts are resolved and how standards are selected, maintained and implemented. Without governance, interoperability may remain a technical aspiration without institutional authority.

This domain examines strategies, standards governance, terminology policies, implementation guides, public accountability and transparent decision-making. It also considers whether governance includes clinicians, patients, laboratories, payers, researchers, software developers and public authorities.

A mature ecosystem does not merely publish standards. It defines how standards are chosen, updated, implemented, monitored and corrected.

4.5 Domain 2 - Technical Interoperability

Technical interoperability concerns the capacity of systems to exchange information through reliable, documented and secure mechanisms. It includes APIs, messaging, data transport, identifiers, infrastructure, network availability and integration.

This domain does not evaluate technology for its own sake. It asks whether technical capacity supports meaningful exchange and whether interfaces, dependencies, testing and failure states are documented.

Database and operating-system quality also matter because unsupported infrastructure, non-exportable data or untested recovery can make interoperability fragile.

4.6 Domain 3 - Identity, Consent and Trust

Identity mechanisms should reliably link information to the appropriate patient, professional and organisation while preserving purpose limitation and accountability. Consent or another lawful basis must be operational rather than merely theoretical.

Trust depends on provenance, secure authentication, access control, auditability and understandable governance. A technically interoperable system that cannot explain who accessed what or how a record was corrected is not mature.

IML favours an identity-access layer that can work with existing national or organisational systems rather than inventing a new universal identity number.

4.7 Domain 4 - Adoption and Use

A capability that exists only in a pilot or procurement catalogue has not yet demonstrated routine maturity. Adoption requires actual use in clinical and organisational workflows.

Training, usability, incentives, professional roles, access rights and workflow design shape whether interoperability becomes useful. IML therefore treats professional role alignment as an important aspect of Adoption and Use.

Adoption evidence should distinguish national availability, local activation and demonstrated routine use.

4.8 Domain 5 - Security and Resilience

Resilience is broader than prevention of attack. Health systems must assume that software defects, network outages, human error, ransomware, supplier failure or physical disruption may occur. The critical question is how quickly essential functions can be restored and whether clinical care can continue safely in degraded mode.

For this reason, backup without restoration testing should not be considered full resilience. Similarly, an incident-response plan that has never been exercised provides weaker evidence than one supported by drills, post-incident analysis and documented corrective action. Security and Learning are therefore naturally connected.

Security and resilience include cybersecurity governance, patching, hardening, segmentation, incident response, access logging, backup, restoration, continuity and the ability to maintain essential care after disruption.

Official claims of security are important evidence, but observed incidents and their operational consequences also matter. Repeated compromise, prolonged loss of availability or failure to restore critical systems can justify a downward correction of a security assessment.

Transparency about incidents should not itself be penalised. The assessment concerns demonstrated security performance and learning from incidents, not the willingness to disclose them.

4.9 Domain 6 - Feedback, Correction and Learning

Learning requires a pathway from observation to durable change. An incident can be documented yet produce no improvement if responsibility is unclear, the corrective action is not implemented or the same problem reappears without review. IML therefore looks beyond reporting volume toward the effectiveness of the feedback loop.

Documentary reliability belongs in this domain because public information is part of institutional memory. When official references disappear, contradict one another or remain obsolete after major changes, users cannot reconstruct what rules applied. Maintaining correctable documentation is therefore an operational form of learning.

A mature ecosystem does not merely exchange information. It detects errors, corrects them, propagates corrections, learns from incidents and modifies institutions or software so that the same failure is less likely to recur.

Feedback transforms interoperability from a static infrastructure into a learning system. Without correction, errors circulate. With correction, information becomes more trustworthy over time.

The September 2026 revision makes explicit that Learning also includes institutional documentary reliability and institutional answerability. Formal digital capacity cannot be considered fully mature if official evidence cannot be maintained or responsible bodies cannot substantively answer documented problems.

4.10 Cross-cutting dimension - Institutional Engagement

Institutional Engagement evaluates whether organisations expose practical collaboration pathways, participate in review and implementation, provide contact points and support shared governance.

INS-9 Institutional responsiveness to external proposals may remain here when it measures the existence of procedures and contact pathways. It does not replace LRN-5, which measures whether responsible institutions actually answer and follow through.

4.11 Cross-cutting dimension - Payer Interoperability

Payer Interoperability evaluates whether public, statutory, mutual or private financing organisations can exchange appropriate information with clinical and public-health actors under trustworthy governance.

Indicators must be adaptable across financing models and must not assume that administrative information is automatically sufficient for clinical interpretation.

4.12 Cross-cutting dimension - AI Readiness

AI readiness is the capacity to deploy AI safely, transparently and usefully. It depends on data quality, context, governance, security, validation, bias monitoring, human oversight, incident reporting, auditability and training.

Access to a powerful model is not maturity. A mature ecosystem must know when not to automate and must preserve provenance when AI is used.

4.13 The IML maturity logic

Different ecosystems may progress in different sequences. Some may begin with governance, others with technical exchange, identity, cybersecurity or a focused clinical use case.

Maturity should be understood as the capacity to learn, adapt and improve rather than possession of a fixed technological architecture.

Each assessment should identify the current profile, the most relevant next step and the evidence required to demonstrate progress.

4.14 Transition to the methodology

Part IV provides the conceptual structure. Part V translates it into indicators, evidence grading, scoring, source assessment, validation and expert review.

Without methodology, concepts remain inspirational. With methodology, they become discussable, testable and improvable.

4.9.1 Learning indicator set LRN-1 to LRN-5

Code	Indicator	Operational meaning
LRN-1	Error detection and reporting	Mechanisms for detecting, reporting and investigating data, workflow or interoperability errors.
LRN-2	Continuous improvement	Use of incidents, audits, outcomes, user feedback and interoperability indicators to change practice or systems.
LRN-3	Traceable correction and reuse	Correction, versioning and traceable reuse of database-held information so that later users can see what changed.
LRN-4	Institutional documentary reliability	Ability of public institutions to maintain official references that remain accessible, precise, current, attributable and correctable over time.
LRN-5	Institutional responsiveness and answerability	Demonstrated capacity to receive, examine and substantively answer relevant documented inquiries and to act on authoritative findings, warnings and recommendations.

4.9.2 Institutional answerability - reference interpretation

Score	Reference interpretation
100	Substantive response or corrective action within 30 days, with traceable follow-through.
75	Substantive response in approximately 31 to 60 days, with usable follow-through.

Score	Reference interpretation
50	Delayed or incomplete response, but sufficiently substantive to support action or review.
25	Acknowledgement, generic answer or formal process without substantive answer or effective follow-through.
0	Persistent non-response, non-enforcement or failure to act after a reasonable opportunity.
Not assessed	Insufficient evidence, issue outside remit, or receipt of the inquiry cannot be established.

Automated acknowledgement is not a substantive response. Authoritative reports by courts of audit, parliamentary bodies, regulators, ombuds institutions or equivalent independent control bodies may directly inform LRN-5 when they document repeated unresolved failures of evaluation, enforcement, correction or follow-through. Individual correspondence should normally remain corroborating evidence and should be privacy-redacted.

Where the evidence supports it, persistent institutional silence or inaction may be described by its functional effect as passive institutional obstruction. The term should not be used to impute personal intent.

Key Messages

- The six-domain framework supports continuous improvement rather than ranking.
- Learning now explicitly includes LRN-1 to LRN-5.
- Formal institutional procedures and demonstrated institutional answerability are different constructs.
- Security and resilience should be informed by observed operational consequences as well as declared controls.
- AI Readiness is cross-cutting and depends on trustworthy information, governance and human oversight.

PART V - Methodology

5.1 Why methodology matters

A maturity framework becomes scientifically useful only when concepts can be translated into transparent indicators and evidence rules. Methodology turns a vision into a process that can be criticised, reproduced and improved.

The purpose is learning, not political ranking. Measurement is the beginning of improvement rather than its conclusion.

5.2 From concepts to indicators

Indicators should be relevant, measurable, comparable, actionable and transparent. An indicator that cannot guide an improvement pathway should be questioned.

Each indicator should identify the phenomenon being measured, evidence types, scope, limitations and scoring logic. Qualitative judgement may remain necessary, but the basis for judgement should be visible.

5.3 Evidence before opinion

Evidence selection should also protect against confirmation bias. Investigators may be tempted to search only for sources that support an initial impression of a country or institution. IML should therefore document contradictory sources and explain why one source is considered more authoritative or more relevant to the indicator.

Search reproducibility becomes increasingly important when AI assists research. Candidate-source discovery can be accelerated by AI, but the final evidence register should contain stable source identifiers, dates, summaries and limitations that a human reviewer can inspect independently of the model that suggested them.

IML should avoid relying exclusively on self-declaration. Policy documents, legislation, technical documentation, implementation reports, audits, scientific publications, public datasets, direct demonstrations and structured expert input provide different kinds of evidence.

The same provisional score may have very different credibility depending on its evidence base. A national strategy should not be treated like observed routine implementation.

Absence of public documentation does not automatically prove absence of capability. It means that evidence is uncertain and should be recorded as such.

5.4 Proposed evidence grading

Early IML versions use an evidence ladder from no identified evidence to mature and reusable evidence. This is an operational grading of documentation and implementation, not a claim of scientific certainty.

Evidence grading should remain separate from performance scoring. Strong evidence may document poor performance, while a high provisional performance score may rest on weak evidence.

5.5 Scoring philosophy

IML scoring reflects progressive maturity rather than binary compliance. Scores should be interpretable at domain and indicator level and should never hide the documentary basis of the judgement.

Where a domain score is revised because a new criterion is adopted, the revision and its rationale should be visible in the profile.

5.6 Country profiles rather than country rankings

A country profile should be read as a snapshot of evidence at a defined date. It does not describe every hospital, clinician or local implementation in the country. National profiles inevitably compress regional variation, and the limitation should be stated explicitly.

Profiles are therefore starting points for dialogue. Local experts should be able to challenge a source, propose stronger evidence or show that a national statement is not representative of routine practice. The methodology should make correction easier rather than defend an initial score.

A profile can show simultaneous strength and weakness: advanced identity infrastructure with poor adoption, strong policy with weak correction, or sophisticated digital services with recurrent security incidents.

Country profiles should display the six-domain shape, strengths, points to watch, sources and limitations. An overall orientation signal may be useful but must not replace the domain table.

Neutral or unassessed status should mean insufficient examination rather than low maturity.

5.7 Weighting and aggregation

Weighting should remain transparent and empirically testable. Equal weighting is easy to understand but may not reflect the clinical consequences of different domains.

IML should avoid over-precision before validation. Domain scores should be accompanied by narrative interpretation, and changes in methodology should be versioned so that time trends remain intelligible.

5.8 Data sources and triangulation

Technical specifications may reveal capability; scientific publications may describe implemented infrastructures; interviews may reveal workflow and readiness; demonstrations may confirm operational capability; routine data may show actual use.

A mature assessment triangulates evidence rather than relying on a single channel. When evidence is missing, uncertain or inaccessible, the assessment should say so explicitly.

Official statements are important sources but should be confronted, where appropriate, with independent audit, documented incidents and implementation evidence.

5.9 Expert review

Health Information Ecosystems are socio-technical systems. Technical experts can evaluate standards and architecture; clinicians can evaluate usefulness and workflow; public-health specialists can evaluate surveillance; legal experts can evaluate governance; patients and civil society can evaluate trust, consent and equity.

Multidisciplinary review reduces the risk that IML becomes technically elegant but clinically weak, clinically aspirational but technically unrealistic, or institutionally blind.

5.10 Validation strategy

Validation should include both construct validity and practical usefulness. It is not enough for experts to agree that an indicator sounds reasonable. The framework should eventually show that different reviewers can apply it consistently and that the resulting recommendations correspond to real implementation bottlenecks.

Pilot validation may also reveal that some indicators are too dependent on public documentation or too difficult to compare across systems. Such findings should lead to revision rather than be treated as failure. A learning framework must apply its own Learning principles to itself.

The methodology should be validated progressively through expert review, comparison with adjacent frameworks and empirical testing in concrete use cases.

The AMR demonstrator provides the first validation pathway. Later validation should examine whether IML assessments identify real bottlenecks and whether recommended actions lead to measurable improvement.

5.11 Versioning and reproducibility

Each version should document changes in definitions, indicators, scoring logic and evidence requirements. Country score changes should be distinguishable from methodology changes.

Version control supports scientific humility. IML should be designed to learn from critique and implementation rather than present early choices as permanent.

5.12 Methodological limitations

Publicly available evidence may be incomplete. Some indicators are easier to measure in high-resource countries. Institutional engagement and answerability are difficult to quantify. Local implementation may exist without national documentation, while national strategy may exist without operational use.

IML should make these limitations visible. A transparent limitation strengthens credibility more than an unjustified claim of certainty.

5.13 Transition to the demonstrator

The methodology creates the bridge between concept and implementation. AMR is the next test because it requires integration of microbiology, clinical context, treatment, outcomes and public-health surveillance.

The sequence remains: Measure -> Understand -> Recommend -> Implement -> Evaluate -> Learn.

5.4.1 Evidence grading template

Grade	Evidence level	Interpretation
E0	No evidence identified	No public or provided evidence supports the indicator.
E1	Statement or intention	A policy statement, strategy or declaration exists, but implementation is not documented.
E2	Partial implementation evidence	Pilot projects, local implementation or limited operational evidence exist.
E3	Documented operational implementation	The capability is implemented and documented in routine practice.
E4	Evaluated implementation	The implementation has been evaluated, audited or described in scientific or institutional reporting.
E5	Mature and reusable evidence	Evidence demonstrates sustained use, governance, monitoring and capacity for reuse or replication.

5.8.1 Authoritative control evidence and Learning

For LRN-5, authoritative control evidence can be primary rather than merely supplementary. Courts of audit, parliamentary bodies, regulators, ombuds institutions or equivalent independent bodies may document repeated failures to evaluate, enforce, correct or follow through. Such findings can demonstrate weak institutional answerability even when an individual response-time dataset does not exist.

The scope of each report must still be stated. A report concerning selected national programmes should not be generalised automatically to every institution or every inquiry.

Key Messages

- The purpose of methodology is continuous improvement, not ranking.
- Evidence grading must remain separate from performance scoring.
- Country profiles should expose uncertainty, sources and limitations.
- Authoritative audit evidence can directly inform Learning when it documents failure to act.
- Versioning is essential to distinguish real change from methodological change.

PART VI - Demonstrator: Antimicrobial Resistance and Multidrug-Resistant Bacteria

6.1 Why AMR/BMR?

Antimicrobial resistance is a global clinical and public-health priority and a particularly strong interoperability test. The relevant information is distributed across microbiology, clinical records, pharmacy, public health and sometimes payer datasets.

IML begins with a focused use case rather than attempting to solve every interoperability problem at once. Multidrug-resistant *E. coli* urinary tract infection provides a common, clinically meaningful pathway.

6.2 From resistant isolate to clinically meaningful infection

A resistant isolate is a laboratory finding. A clinically meaningful infection is an interpretation that depends on symptoms, specimen, bacterial count, host context, diagnosis and treatment.

Conflating the two can distort surveillance and clinical decision support. The demonstrator therefore tests whether interoperability preserves the distinction between biological detection and clinical significance.

6.3 Colonisation, contamination, asymptomatic bacteriuria and symptomatic infection

The distinction is clinically important because inappropriate treatment of colonisation or asymptomatic bacteriuria can contribute to antimicrobial resistance and expose patients to unnecessary adverse effects. Conversely, failing to recognise a clinically significant infection can delay effective treatment. The interoperability model should therefore avoid reducing a culture result to a binary resistant/not-resistant flag.

Temporal relationships matter. Symptoms may precede the culture, antibiotics may have been started before sampling, and the final diagnosis may change after results are reviewed. The minimum dataset should preserve enough chronology to support interpretation rather than assuming that all data elements are contemporaneous.

The same organism can represent different states. Colonisation may be clinically important without active infection; asymptomatic bacteriuria may not require treatment in many contexts; contamination may invalidate a result; symptomatic infection may require prompt treatment.

Structured data should support these distinctions without forcing certainty where the clinical record remains ambiguous.

AI-assisted classification may help identify missing fields but must not invent a diagnosis.

6.4 The urinary tract infection use case

UTI is suitable because symptoms, urine testing, culture, bacterial count, susceptibility, treatment and outcome can be represented in a manageable minimum dataset.

The use case also illustrates the need to connect primary care, laboratory systems and sometimes hospital or pharmacy information.

6.5 Minimum clinical dataset

The minimum dataset should include symptoms, fever where relevant, complication context, leucocyturia or urinalysis findings when available, specimen information, culture result, bacterial count, organism, susceptibility profile, final clinical interpretation, treatment and evolution.

The dataset should remain deliberately small enough for real implementation while preserving the variables required to distinguish meaningful clinical states.

6.6 Microbiology and antibiogram data

Microbiology requires semantic precision: organism, specimen, method, concentration or count, susceptibility categories, antimicrobial agents and laboratory standards may all affect interpretation.

LOINC, organism terminologies and standard antimicrobial vocabularies can support exchange, but local mappings must preserve source values and version information.

6.7 Treatment and outcome

Resistance only becomes clinically useful when linked to what clinicians did and what happened next. Treatment choice, dose, route, timing, changes and outcome are therefore part of the demonstrator.

Outcome may include symptom resolution, recurrence, admission, complications or treatment modification. Even partial follow-up can be useful if missingness is explicit.

6.8 Public health surveillance

Public-health surveillance needs scalable aggregate indicators, but aggregation should not erase clinical meaning. Counts of resistant isolates and counts of symptomatic infections answer different questions.

The demonstrator aims to show how a clinical context layer can improve the interpretation of surveillance without requiring public-health systems to become full clinical records.

6.9 What interoperability changes

Interoperability can reduce duplicate data entry, make laboratory information actionable in context, support correction, and create a reusable evidence trail for quality improvement.

It can also reveal what remains unknown. A mature system should distinguish absence of a symptom from absence of documentation.

6.10 Pilot proposal

A pilot should define an explicit evaluation plan before implementation. Useful measures may include completeness of clinical context, time required to interpret imported results, frequency of duplicate entry, number of corrected records, proportion of cases that can be classified confidently and professional usability.

The pilot should also record failures. A connector that cannot parse a real laboratory correction, a terminology mapping that proves ambiguous or a clinician who rejects an AI suggestion are scientifically useful observations. The goal is not to demonstrate success at all costs; it is to learn which architecture survives real use.

A pilot should begin with a bounded clinical pathway, defined participating organisations, a minimal dataset, source-preserving import, terminology mapping, explicit governance and measurable outcomes.

The first success criterion should not be message volume. It should be whether the integrated information improves interpretation, reduces ambiguity or supports better review.

AI-EVIDENCE-01 and AI-EVIDENCE-02A can later be tested against AMR questions, for example by retrieving evidence and translating traceable claims into clinically relevant context while preserving the evidence trail.

6.5.1 Reference minimum dataset

Category	Minimum elements
Clinical presentation	Urinary symptoms, fever/systemic signs, duration, pregnancy or other relevant complication context.

Category	Minimum elements
Urinalysis	Leucocyturia and other available urine findings, with method where material.
Microbiology	Specimen, collection date, organism, bacterial count/quantitation, susceptibility profile, method/standard.
Interpretation	Contamination, colonisation, asymptomatic bacteriuria, symptomatic infection, uncertain/other.
Treatment	Antimicrobial, dose/route where relevant, start/change/stop, rationale where available.
Outcome	Improvement, persistence, recurrence, complication, admission, follow-up unknown.
Provenance	Source organisation/system, timestamps, correction/version status.

Key Messages

- Resistant isolate is not synonymous with infection.
- Clinical context is required for meaningful AMR surveillance.
- A minimum dataset should be implementable and still preserve decisive distinctions.
- Treatment and outcome close the loop from microbiology to health effect.
- AMR provides a practical validation pathway for both interoperability and evidence-aware AI.

PART VII - From Assessment to Action

7.1 Why assessment is not enough

An assessment has value only if it leads to a better understanding of what should change. IML therefore links each maturity observation to potential actions, evidence needs and implementation priorities.

Recommendations should be proportionate to context rather than assuming that every ecosystem should deploy the same technology.

7.2 Recommendations by maturity profile

An ecosystem with weak governance may need responsibility and standards stewardship before complex integration. An ecosystem with strong infrastructure but weak adoption may need workflow redesign and professional support. An ecosystem with recurrent incidents may need resilience and correction before expansion.

The six-domain profile therefore acts as a decision map rather than a grade.

7.3 Supporting low-resource ecosystems

Low-resource does not mean low ambition. It means that architecture must respect constraints such as intermittent connectivity, limited technical staff, older hardware, variable electricity supply and restricted procurement budgets. Systems that require continuous cloud access or expensive proprietary licences may be inappropriate even when technically sophisticated.

Open-source software can help only when documentation, installation, updates and support are realistic. IML therefore links low-resource deployment to university partnerships, reproducible packaging and the capacity to run locally. Sustainability matters more than the novelty of the technology.

Low-resource settings may benefit from open-source components, local deployment, offline-capable workflows, simple structured exchange and university-supported implementation.

IML should avoid recommending architectures whose ongoing maintenance cost exceeds local capacity. A useful first installation is better than an ambitious platform that cannot be sustained.

The Guatemala adaptation is intended to test this principle in practice.

7.4 Supporting high-resource fragmented systems

High expenditure and sophisticated national services do not guarantee continuity. Fragmentation can persist through proprietary silos, complex identity arrangements, delayed implementation or inconsistent institutional follow-through.

High-resource systems therefore require the same evidence discipline as low-resource ones. Formal availability should not be confused with routine clinical usefulness.

7.5 The French paradox

The French case is also useful methodologically because it demonstrates why official descriptions and observed outcomes should be analysed together. A country can have strong legal frameworks, national services and major investment while also experiencing delayed implementation, interoperability gaps or recurrent cyber disruption. Both sides of the evidence belong in the profile.

The purpose is not to single out France, but to demonstrate that maturity cannot be inferred from institutional sophistication. Similar analyses should be applied to every country when authoritative audit or incident evidence is available, while taking care not to penalise systems merely because they are more transparent about their difficulties.

France illustrates the possibility of advanced national digital-health programmes alongside persistent interoperability and implementation challenges. The analytical value of the case lies precisely in avoiding a simplistic label of either success or failure. Official strategies and infrastructure provide important evidence, while authoritative audits and documented operational incidents can reveal gaps between declared capability and demonstrated performance.

IML therefore uses the French case to show why Learning, documentary reliability, institutional answerability and security outcomes belong in maturity assessment.

7.6 The Danish reference without idealisation

Denmark provides important references in microbiology and surveillance, including national infrastructures such as MiBa, DANMAP and HAIBA. These systems demonstrate the value of linking laboratory and population data. IML uses Denmark as a comparative reference without assuming that every clinically relevant variable is always available in structured form at the moment of care. A reference becomes more useful, not less, when its limitations are explicit.

7.7 Institutional inertia, engagement and answerability

Interoperability can stall even when technical solutions are known. Competing priorities, procurement, legal caution, workforce limitations and institutional inertia can delay action.

IML distinguishes understandable delay from persistent failure to respond or act on authoritative findings.

LRN-5 provides a transparent place to record demonstrated institutional answerability.

Individual correspondence should not become a personal grievance metric. The unit of analysis is the institution's demonstrated capacity to process relevant evidence and follow through.

7.8 Community building

IML should invite clinicians, patients, informaticians, microbiologists, public-health specialists, payers, security experts, legal experts, universities and open-source contributors.

Community building should invite critique as well as collaboration. A framework that cannot absorb disagreement is not a learning framework.

7.9 From pilot to reusable service

Focused pilots should be designed so that their useful components can be reused: identity, consent, terminology, import, audit, correction, analytics and evidence retrieval.

The aim is cumulative infrastructure rather than a succession of isolated demonstrations.

7.10 Software-quality benchmarking and contingency

Open-source availability alone does not guarantee quality. IML should evaluate installation reproducibility, lifecycle support, dependency management, export, backup, recovery, documentation, cybersecurity and clinical safety.

Health systems should understand contingency before a vendor or project disappears. A reference open-source module can be valuable as a fallback only if it is actually installable, testable and maintainable.

Key Messages

- Assessment should lead to specific, proportionate action.
- Low-resource and high-resource ecosystems can fail for different reasons.
- Reference countries should be used without idealisation.
- Institutional answerability is part of learning, not merely etiquette.
- Open-source reference software should be tested as real contingency infrastructure.

PART VIII - Open Clinical Workspace

8.1 Why an open clinical workspace?

IML requires an implementation bridge between assessment and practical improvement. The Open Clinical Workspace provides a vendor-neutral, open-source-oriented environment in which clinical information can be imported, contextualised, corrected, audited and reused.

The workspace is not intended to become another closed national platform. Its scientific value lies partly in inspectability and the possibility of replacing components.

8.2 Not replacing existing systems

Hospitals, laboratories, pharmacies and practices already use specialised systems. Replacing all of them is unrealistic and often undesirable.

The workspace should therefore coexist with existing software, retrieving or receiving relevant information and returning structured outputs through explicit interfaces.

8.3 Interacting with existing software

Integration can begin with files, messages or APIs depending on local maturity. The important design property is an explicit connector contract that preserves source provenance and isolates national or vendor-specific assumptions from the common clinical model.

A connector should fail visibly rather than silently transform information.

8.4 Open standards

IML should use mature open standards where they fit the problem. Standards improve portability and shared understanding when implemented consistently.

Standards should not become an excuse for unnecessary complexity. A bounded local interface may be appropriate during early implementation if it remains documented and replaceable.

8.5 FHIR, openEHR and terminology services

FHIR provides a practical resource-oriented interoperability ecosystem; openEHR provides important approaches to durable clinical modelling; terminology services support semantic consistency. IML can learn from all three without declaring one tool sufficient for every layer.

The reference architecture should make these components interoperable rather than inseparable.

8.6 PostgreSQL and modular architecture

The current PostgreSQL work uses explicit schemas to separate clinical, identity, terminology, AI and system concerns. Such separation improves readability and can support least-privilege access, but schema boundaries alone are not security controls. Roles, grants, application-layer validation and deployment configuration remain necessary.

Migration discipline is equally important. Recent terminology canonicalisation work demonstrates a conservative approach in which legacy schemas are preserved, destructive operations are avoided and migrations abort when unexpected content differences are detected. This pattern is valuable for clinical systems because a migration should fail safely rather than silently discard information.

PostgreSQL is the current reference relational data foundation for the workspace. It supports constraints, transactions, structured JSON metadata, migration discipline, audit patterns, backup, recovery and transparent inspection.

The choice is a reference implementation decision, not a claim that PostgreSQL is the only acceptable database. Exportability and documented schemas are more important than technological loyalty.

Modules are separated conceptually and, where useful, through database schemas so that clinical, identity, terminology, AI and system concerns can evolve without one opaque data store.

8.7 Clinical context layer

The clinical context layer preserves the meaning surrounding observations. It links results to symptoms, diagnosis, indication, treatment, evolution and relevant risk factors.

This layer is especially important for AMR, where the same microbiological result can support different interpretations.

8.8 Patient identity and consent

The workspace should work with existing national or organisational identity systems. It should support patient and professional identifiers, provenance, lawful access, consent or other appropriate legal basis, and auditable contextual authorisation.

It should not create a new universal personal identity number.

8.9 Import on demand

Import on demand supports proportionality. A workspace should retrieve the information required for a defined clinical or research purpose rather than replicating every available record by default.

Local caching, synchronisation and offline strategies may still be necessary, but the architecture should make the purpose and lifetime of replicated information explicit.

8.10 AI support under human responsibility

Local structured extraction is a practical example of this philosophy. Dictation or observation can be transformed into candidate facts, but the candidate remains separate from the validated clinical record until reviewed. This keeps the speed advantage of AI while preserving the clinician as the accountable author of the record.

The same pattern can be reused for terminology suggestions, evidence summaries and missing-context alerts. AI can prepare the next action, but the system should make the transition from machine candidate to human-validated information explicit and auditable.

AI may assist structured extraction from dictation or observation, evidence retrieval, terminology suggestions, missing-context detection, summarisation and pattern recognition.

Candidate facts or suggestions must remain subject to clinician review where they affect the record. AI output should never silently overwrite authoritative clinical information.

8.11 Minimum viable prototype

A useful first prototype should demonstrate patient/encounter context, terminology, laboratory import, provenance, audit, correction and a focused clinical workflow. It should be installable and testable without requiring an entire national architecture.

Success should be measured by clinical usefulness and reproducibility rather than feature count.

8.12 Database and operating-system quality

Database quality includes integrity, security, traceability, reversibility, portability, correction, backup, restoration and long-term maintainability. Operating-system quality includes supported lifecycle, patching, hardening, portability and recoverability.

These properties belong to interoperability because a system that cannot be maintained or migrated cannot reliably preserve knowledge.

8.13 From prototype to reusable service

If the initial prototype proves useful, the same modules can support vaccination continuity, chronic disease, discharge coordination, medication safety, registries, cancer pathways or emergency summaries.

Each new use case should strengthen common infrastructure rather than create a new isolated application.

8.14 Governance of the Workspace

The workspace should be governed by clinicians, patients, microbiologists, public-health representatives, payers, informaticians, security experts, legal experts and open-source contributors.

Governance should define priorities, approve data models, supervise security, document decisions, maintain issue tracking and ensure that the software remains aligned with health objectives.

Open-source transparency is not enough. Trust requires maintenance, testing, correction and accountable stewardship.

8.15 Transition to Responsible AI and Scientific Integrity

More connected and contextualised information increases the potential usefulness of AI. It also increases responsibility because AI can amplify both good information and hidden errors.

The next part therefore treats AI as an assistive layer governed by scientific integrity, provenance and human responsibility.

8.16 Proposed Modules of the Open Clinical Workspace

The original module proposal remains: data import, terminology service, clinical context layer, identity and consent, audit log, analytics, database/storage quality, AI assistance, export and a documented open-source reference toolchain.

The September 2026 software workstream now makes several of these modules more concrete without converting them into validated clinical services.

8.16.1 Current reference architecture - September 2026

Layer	Current reference direction
Clinical core	Patient identity/context, encounters, diagnostic assessments, observations, results, clinical history, correction and provenance.
Terminology	Canonical terminology structures; KSH97-P encounter reasons; progressive ICD-10 mapping; translation review workflow; ICD-11 readiness remains scaffolded.
Open integration layer	Explicit connectors for laboratories, imaging, registries, identity, consent and national services.
Country packs	Common core with national language, terminology, identity, billing, messaging and regulatory adapters.
Data foundation	PostgreSQL reference architecture with local or remote database/API patterns.

Layer	Current reference direction
AI service layer	Local structured clinical extraction plus evidence retrieval; consequential output remains subject to human validation.
Operations	macOS, Windows and Linux targets; Debian reference server baseline; Ubuntu LTS secondary; offline-capable workflows and local peripherals anticipated.

8.16.2 France reference pack

The France pack is a real country adaptation rather than a translation. Its current scope includes INS, MSSante, DMP/Mon espace sante, CPS/e-CPS, Sesam-Vitale, SCOR, Segur-related constraints, prescribing requirements, NGAP, CCAM, CIM-10-FR, LOINC FR, ALD, C2S, AME, AT/MP, laboratory and imaging integration.

The first controlled country-specific integration target is HPRIM laboratory results. The objective is to preserve original provenance while normalising clinically useful content into the common model.

The pack should remain separable from the common core so that French national services can evolve without forcing the core clinical model to change.

8.16.3 Guatemala reference pack

Guatemala is the second reference adaptation. The first software task is a Spanish-language version of the current workspace, while preserving the common clinical model.

Intermittent connectivity, limited infrastructure, local deployment and maintainability are first-class design constraints. The country-pack strategy is intended to make later extension additive rather than create a separate application for each country.

8.16.4 Terminology workstream

The terminology work now treats `iml_terminology` as the canonical active terminology environment and preserves legacy schemas when canonicalisation occurs. KSH97-P encounter reasons, a terminology bridge and a translation review workflow are active components of the current module registry.

The translation workflow is deliberately review-based. French, Latin American Spanish and later Guatemala-specific designations should not be automatically promoted as official terminology labels merely because an AI or translation model produced them.

ICD-11 readiness is scaffolded as a future multi-classification capability and shadow-coding environment. No live ICD-11 clinical coding should be claimed until validation is complete.

8.16.5 Local structured clinical extraction

The current module registry also includes local structured clinical extraction. Its role is to identify candidate facts from dictation and observation for clinician review. This architecture is intentionally conservative: extraction creates candidates, not silent chart truth.

Local processing can reduce unnecessary external disclosure and improve resilience, but local execution does not eliminate the need for model validation, auditability, user-interface clarity and correction.

8.16.6 A common core with replaceable components

The Open Clinical Workspace should not become vendor lock-in created in the name of open source. Clinical data, terminology, interfaces, evidence retrieval, analytics and AI services should remain separated through documented contracts and explicit provenance.

A database host, model provider, terminology server or national connector may change without destroying the history or meaning of the underlying clinical record. Reversibility is therefore both a software-quality property and a health-information governance principle.

8.16.7 Deployment and resilience

Offline capability creates design questions that must be resolved explicitly: which data are cached, how conflicts are detected, how identifiers are reconciled, how audit logs are merged and what happens when two sites modify the same information before reconnection. These are not edge cases in constrained environments; they are central operational requirements.

Multiplatform support should also be tested through real packaging and peripheral integration. A theoretical claim that software runs on Windows, macOS and Linux is weaker evidence than reproducible installers, automated tests and documented use of card readers or laboratory interfaces on those platforms.

The workspace is designed for local or remote PostgreSQL configurations, API access, progressive synchronisation and offline-capable operation. A site should be able to adopt useful modules without purchasing an entire national architecture.

Reference deployment is multiplatform. macOS and Windows matter for professional workstations; Linux provides a conservative server environment, with Debian as the reference baseline and Ubuntu LTS as a secondary target. Local health-card readers and other peripherals should be anticipated rather than excluded by a cloud-only design.

8.16.8 Clinical knowledge and medication decision support

The Open Clinical Workspace should not become a repository of medical publications. Its intended role is to read or receive information from lawful sources, extract clinically useful claims, preserve the source trail and represent the circumstances in which those claims were established. Publications, guidelines, regulatory documents and clinical references remain evidence sources rather than the final operational object.

A clinically useful claim should remain linked to provenance, the population actually studied or targeted, relevant outcomes, certainty or methodological qualification where available, jurisdiction, dates and limitations. At consultation time, the workspace can then compare these patient-independent objects with the local clinical context and show the clinician why a claim appears applicable, only partially applicable or not directly applicable.

Four dimensions must remain separate: authority for action, operational ease of retrieval, scientific certainty, and applicability to the person. A national regulatory rule may have priority for action in its jurisdiction without constituting the strongest scientific evidence. Conversely, a high-certainty international recommendation may require caution when the patient differs materially from the population on which the recommendation rests.

Country / layer	Example sources	Intended IML role
France - regulatory	Applicable RCP/SmPC, ANSM/EMA and official national product information	Authority and regulatory provenance for medication action.
France - operational	VIDAL or other authorised/licensed medication reference	Rapid structured retrieval and prescription support; not a substitute for regulatory authority.
Sweden - regulatory	Läkemedelsverket product information	Official Swedish regulatory product information.
Sweden - operational	FASS	Widely used practical medication information and product-summary presentation.
Sweden - decision support	Janusmed Interaktioner and Riskprofil	Interaction and multi-drug pharmacodynamic risk support.
Sweden - clinical reference	Läkemedelsboken	Practical treatment and medication reference, particularly relevant to primary care.
Sweden - national guidance	Socialstyrelsen	National guidance, priorities and implementation context.

These examples are architectural roles, not endorsements or claims of unrestricted access. Commercial, institutional and public sources require connector-specific technical validation, licensing or reuse analysis. Where a source can be consulted but not lawfully ingested or persistently stored, IML should preserve only the permitted metadata, link-out and transient processing state.

Medication reasoning should operate at the active-ingredient level. Fixed-dose combinations should be decomposed into their component substances so that duplicate therapy, pairwise interactions and cumulative effects are visible. Cumulative risk may involve several medicines together with renal function, age, laboratory values or comorbidity; clinically important examples include hyperkalaemia, hypotension, bradycardia, bleeding, QT prolongation, hypoglycaemia, sedation/falls, anticholinergic burden and kidney injury.

The consultation interface should compress this prepared knowledge rather than perform a full literature review while the patient waits. High-priority blocking issues, monitoring or dose-adjustment needs, and overall therapeutic coherence should be visible in seconds, with a "Why?" path back to the source. AI may prioritise and contextualise these signals, but the prescriber remains responsible for the decision.

Key Messages

- The Open Clinical Workspace is an implementation bridge between maturity assessment and practical action.
- It should interact with existing systems rather than replace them wholesale.
- The current workstream includes real PostgreSQL, terminology, clinical extraction and country-pack implementation objects.
- France and Guatemala are the first reference country adaptations.
- AI services should remain replaceable, provenance-aware and subordinate to human clinical responsibility.
- Offline capability, portability and contingency are part of resilience rather than optional extras.

- Clinical knowledge should be extracted and traced to sources rather than accumulated as an opaque publication repository.
- Medication support should distinguish regulatory authority, operational references, scientific certainty and patient applicability, and should reason across all active ingredients in combination products.

PART IX - Responsible Artificial Intelligence and Scientific Integrity

9.1 AI as editorial and analytical support

AI can assist drafting, structuring, source discovery, classification and analytical exploration. These functions can increase productivity but also create risks of confident error, incomplete sourcing or hidden transformation.

IML therefore treats AI assistance as something to disclose and audit where it materially contributes to scientific or clinical work.

9.2 Ethical aspiration and public benefit

The project has been inspired by the aspiration that advanced AI should benefit humanity. This is not a claim of endorsement, partnership or institutional affiliation.

Within IML, public benefit means trustworthy information, better decisions, improved prevention, scientific collaboration and more equitable access, without increasing opacity, dependency or fragmentation.

9.3 Artificial intelligence as an ecosystem actor

AI should be modelled as an actor with inputs, outputs, dependencies, version, purpose and accountable human users. It is not an autonomous moral or clinical authority.

Because model outputs influence human attention, provenance and failure modes must be visible.

9.4 Human responsibility

A model may identify correlations or suggest a recommendation; a clinician must interpret their meaning for a specific patient. An institution must determine whether and how the tool may be used.

Human responsibility is therefore a non-negotiable condition of AI use in IML. Consequential outputs require professional review appropriate to risk.

9.5 Transparency of AI-assisted writing and analysis

This manuscript was prepared with editorial and analytical assistance from OpenAI ChatGPT. AI supported drafting, structuring, language refinement and conceptual exploration. It did not replace authorship, accountability or final judgement.

Future IML publications should continue to disclose meaningful AI assistance when it contributes substantially to writing, analysis or interpretation.

9.6 Inspiration from the OpenAI Charter

An ethical aspiration toward broad human benefit is useful only when translated into concrete safeguards. IML interprets it through transparency, proportionality, independence from a single provider, correction and public-interest orientation.

Reference to the Charter should not be read as endorsement of IML by OpenAI.

9.7 Privacy, proportionality and data protection

Health data are highly sensitive. Responsible AI requires privacy by design, data minimisation, purpose limitation, access control, auditability and compliance with applicable law.

The objective is not to feed AI systems every available record. The least necessary information should be used for a clearly justified purpose. External evidence-search expressions should be de-identified.

9.8 Bias, equity and representativeness

AI systems can reproduce inequalities when trained or deployed on incomplete or biased data. Vulnerable populations may be precisely those whose information is most fragmented or least represented.

AI readiness should therefore include bias evaluation, documented limitations, performance monitoring across relevant populations and the ability to suspend use when performance is unsafe.

9.9 Scientific integrity and reproducibility

Reproducibility does not require every reviewer to reach the same interpretation, but it does require enough transparency to understand how the interpretation was produced. Queries, eligibility criteria, source versions, exclusions and transformations should therefore be available at an appropriate level of detail.

When AI participates, additional provenance is required because model behaviour can change between versions. A later reviewer should know which model or tool version contributed, what input set it received, what template or instructions shaped the task and where human modification occurred.

AI-assisted analysis must remain compatible with scientific integrity. Data sources, methods, software/model versions, inclusion and exclusion choices and transformations should be documented sufficiently for review.

AI-assisted country research may identify candidate sources, organise evidence, detect omissions or draft summaries, but it must not infer successful national implementation from the existence of laws or platforms and must not publish a profile as validated without human review.

The same principle applies to clinical evidence: a fluent synthesis is not a substitute for a transparent evidence process.

9.9.1 AI-EVIDENCE-01 - Traceable Clinical Evidence Retrieval

The retrieval layer also enables negative evidence about the search process. If one source is unavailable while others respond, the final user can see that the search was incomplete. If a query returns no results, the system can distinguish a genuine empty result from a failed connector. This distinction is essential before any AI model is allowed to interpret the evidence landscape.

Canonical identifiers support deduplication across sources. The same trial or publication may appear in PubMed, Europe PMC and Crossref. AI-EVIDENCE-01 should preserve source-specific retrieval events while linking them to a shared canonical record when the identifiers justify that conclusion. Title similarity alone should not silently merge records.

AI-EVIDENCE-01 is the current retrieval and provenance foundation for bibliographic evidence in the Open Clinical Workspace. It is implemented as a PostgreSQL scaffold in the `iml_ai` schema and is explicitly not presented as a validated clinical service.

Its central principle is separation. A clinical encounter or diagnostic assessment may be associated with an evidence question, but the external search expression must be de-identified. The search process, source runs, retrieved records and persistent identifiers are stored independently of any generative model.

The source registry currently includes PubMed/NCBI, Europe PMC, ClinicalTrials.gov, Crossref and public Cochrane metadata. Each source serves a different role: biomedical bibliographic discovery, open-access and metadata discovery, trial registration, DOI metadata, and systematic-review discovery.

Each source execution can record success, no results, partial response, unavailability, rate limiting, failure or cancellation. This prevents a later AI system from confusing absence of evidence with technical failure.

Retrieved records are represented canonically and may retain PMID, PMCID, DOI, NCT, Cochrane or other identifiers, as well as publication metadata, abstract availability, open-access state and retraction state.

Publisher full text is not stored by default. Credentials, passwords, session cookies and institutional secrets must not be stored in evidence tables. Access rights and licences remain separate from the bibliographic provenance layer.

Connectors remain disabled until validated at application level. A database schema is evidence of architectural progress, not proof of safe clinical operation.

9.9.2 AI-EVIDENCE-02A - Clinical Evidence Provenance, Applicability and Medication Support

AI-EVIDENCE-02A is proposed as a change of unit from publication-centred retrieval to patient-independent clinical knowledge. The operational sequence is source -> clinically useful claim -> provenance -> studied or targeted population -> outcomes and certainty -> applicability criteria -> patient context -> clinician-facing support. IML does not claim authorship of the underlying medical knowledge.

GRADE should be used as an established methodology when GRADE assessments are available. Certainty is preferably represented at outcome level, with reasons such as risk of bias, inconsistency, indirectness and imprecision kept visible where supplied. IML should not manufacture an authoritative GRADE rating merely because a model can generate one, and recommendation strength should remain distinct from certainty.

MAGIC Evidence and MAGICapp illustrate a complementary approach in which guideline content, PICO questions, evidence profiles, recommendations and links to underlying evidence are digitally structured. Where such objects can be lawfully and technically imported, IML should preserve their identifiers, versions and source relationships rather than ask AI to reconstruct information that already exists in a structured form.

EMA contributes two different layers. Electronic Product Information (ePI) and the SmPC provide authorised medicine information suitable for structured regulatory use, while EMA Clinical Data Publication can expose deeper provenance such as clinical study reports, protocols and statistical-method documentation. These layers can help distinguish what is authorised for use from how the evidence supporting an authorisation was generated.

National sources should be considered first when they are legally or regulatorily applicable to the clinical action. This ordering is a rule of application, not a declaration of scientific superiority. The architecture therefore separates authority/binding status, scientific certainty, operational retrieval priority and patient applicability.

Clinical-trial evidence should retain the population actually studied, inclusion and exclusion criteria, age range, renal or hepatic context, comorbidity, concomitant treatment, duration of follow-up and clinically important outcomes when those data are available. Industry sponsorship does not invalidate a study automatically, but sponsor identity, sponsor role, registration, protocol availability and access to more complete regulatory evidence are relevant provenance and should not disappear from the evidence chain.

Systematic reviews and meta-analyses should not receive automatic superiority merely because results have been statistically pooled. IML should preserve study selection, population and intervention heterogeneity, outcome definitions, publication-bias concerns, indirectness and other limitations. Greater numerical precision cannot by itself repair poor external validity or selective evidence.

Patient applicability is therefore a separate clinical dimension. A high-certainty result in a selected population may be only indirectly applicable to an older patient with renal impairment, multimorbidity and polypharmacy. The future runtime should make that distance visible rather than convert a population-average conclusion into a silent patient-specific instruction.

Medication Decision Support is the first concrete translation target. Knowledge should be prepared and versioned outside the time pressure of a consultation. During care, fixed-dose combinations are decomposed into active substances, contraindications and dose constraints are checked, pairwise and cumulative medication risks are

considered, and monitoring needs are prioritised. Patient-specific resolution should remain in the local clinical environment unless an explicitly governed deployment authorises another architecture.

Layer	IML role	Human-control principle
Source policy	Identify authority, jurisdiction, access rights and operational retrieval route.	No source is promoted simply because it is easy to access.
Clinical claim	Store a concise, normalized clinically useful statement linked to sources.	Verbatim source text is not the default knowledge store.
Provenance	Preserve source identifiers, sponsor context, versions, protocols and transformations.	Clinician/reviewer can trace why a statement exists.
Population & applicability	Represent whom the evidence studied or targeted and where it may not generalise.	Applicability to a patient remains reviewable.
GRADE / structured evidence	Reuse established certainty and PICO structures where available.	No automatic authoritative grading.
Medication reasoning	Decompose combinations and evaluate interaction, cumulative risk, dose and monitoring rules.	Prescriber validates the action.
Consultation output	Prioritise a small number of actionable signals with source drill-down.	AI proposes; clinician accepts, rejects or modifies.

AI-EVIDENCE-02A remains a proposed reference architecture and migration draft in this edition. Schema design is not evidence of clinical safety. Validation requires fixed test cases, source-policy review, clinical usability testing, failure-state testing and prospective assessment before patient-facing recommendations can be claimed.

9.10 AI readiness as a maturity dimension

IML proposes AI Readiness as a cross-cutting maturity dimension. Readiness includes governance, clinical validation, audit mechanisms, bias monitoring, human oversight, transparency, incident reporting, cybersecurity, professional training and the capacity to stop or correct a system.

AI-EVIDENCE-01 and AI-EVIDENCE-02A illustrate why data, evidence provenance and patient applicability should be part of readiness. A model provider can change; the evidence trail should remain.

9.11 Boundaries of AI in health information ecosystems

Not every task should be automated. Not every dataset should be reused. Not every prediction should become a recommendation.

The ability to say no, limit scope, require human review and disable a component is part of maturity. IML should prefer architectures in which the clinical workspace remains usable when AI is unavailable.

9.12 Transition to roadmap and future work

Responsible AI reinforces the central logic of IML. Health is the objective. Information is the foundation. Interoperability is the path. AI may assist this path only if it remains governed by human responsibility, scientific evidence and public trust.

The roadmap must therefore advance software implementation and evidence methodology together.

Key Messages

- AI is an ecosystem actor, not an autonomous clinical or moral authority.
- AI-EVIDENCE-01 preserves a traceable chain from evidence query to source execution and retrieved record.
- AI-EVIDENCE-02A proposes a traceable path from source to clinically useful claim, studied population, certainty, applicability and medication support.
- GRADE, MAGIC and regulatory product information should be reused transparently where available; AI must not silently replace their methods or authority.
- Clinical recommendations must keep authority for action, scientific certainty and patient applicability as separate dimensions, under human validation.

PART X - Roadmap and Future Work

10.1 From manuscript to working programme

IML should continue in parallel as a scientific framework, assessment methodology, country evidence programme, reference software implementation and open collaboration project.

Progress should be visible through versioned documents, reproducible software migrations, explicit module status and validated or clearly provisional country profiles.

10.2 Versioning as scientific discipline

Definitions, indicators, country profiles, database schemas, terminology releases, AI prompts/templates and synthesis outputs should be versioned.

Versioning allows disagreement and correction without erasing history. It is a scientific requirement as much as a software practice.

10.2.1 From specification to reference implementation

The reference implementation should function as a scientific instrument. Its schemas, migrations and interfaces make assumptions concrete enough to test. A vague principle such as correction becomes a versioned data operation; a principle such as provenance becomes explicit identifiers and source records; a principle such as replaceable AI becomes a service boundary that can be disabled.

This translation from principle to code is valuable even before clinical deployment. It exposes contradictions that prose alone can hide and gives external reviewers something reproducible to challenge. The manuscript and software should therefore evolve together while retaining independent version histories.

The transition has begun. Current work includes a PostgreSQL-oriented clinical architecture, canonical terminology structures, local structured clinical extraction, a traceable evidence-retrieval scaffold, an open integration strategy and country adaptation packs.

These implementation objects allow the principles of the manuscript to be tested. They do not by themselves demonstrate clinical safety, effectiveness, regulatory conformity or successful deployment.

AI-EVIDENCE-02A should follow the same discipline: define source-policy, provenance, clinical-claim and applicability objects first; test the medication and evidence flows against fixed cases; validate human review and alert-management workflows; and only then consider clinical-facing use.

10.3 External scientific review

IML should invite critique from clinicians, informaticians, public-health specialists, microbiologists, methodologists, patients, payers, security experts, legal experts and software engineers.

Review should be documented, including disagreements that lead to no immediate change.

10.4 Methodology handbook

A separate methodology handbook should define indicator operationalisation, evidence rules, scoring, source quality, country review procedures, Learning criteria and validation methods in greater detail than the Founding Manuscript.

The Founding Manuscript should retain the scientific rationale while the handbook can evolve more frequently.

10.5 Technical handbook

A companion technical handbook should document database schemas, migrations, APIs, connectors, deployment patterns, terminology services, audit architecture, AI-EVIDENCE-01, AI-EVIDENCE-02A and the Medication Decision Engine.

Implementation documentation should include status labels so that scaffold, active module, reference implementation and clinically validated service are never confused.

10.6 Case studies

Case studies should test IML against real information pathways. AMR remains the primary demonstrator, but vaccination continuity, chronic disease, discharge, medication safety and other use cases can test reusability.

Country case studies should combine public evidence with local expert review.

10.7 Implementation pathway

The next software phase should include reproducible migration testing, simulated or appropriately governed clinical datasets, terminology validation, HPRIM laboratory integration, cybersecurity review, offline/resilience testing, country-pack validation, usability and clinical safety assessment.

AI evidence connectors require contract tests, de-identification tests, failure-state handling, deduplication and source-policy compliance.

10.8 Publication, DOI and open documentation

Scientific versions should be citable and archived. Major conceptual and methodological changes should have stable dates or versions, while software releases should carry independent technical versioning.

Open documentation should make the project inspectable without implying endorsement by referenced organisations.

10.9 Community building

IML should grow through partnerships with universities, health institutions, public-health agencies, payers, open-source communities and professionals willing to test or critique the framework.

The project should remain able to receive negative evidence and revise its assumptions.

10.10 Future research agenda

Future research should also investigate the organisational consequences of open reference software. Does a vendor-neutral fallback improve negotiating power, continuity planning or local innovation? Does the existence of an inspectable reference implementation make institutional claims easier to verify? These questions connect software architecture to governance and resilience.

For AI-EVIDENCE-02A, prospective studies should compare machine-assisted and conventional evidence-to-clinical-knowledge workflows on fixed source sets. Outcomes should include missed or misrepresented evidence, extraction accuracy, reviewer time, disagreement, reproducibility, provenance completeness, applicability assessment and the effect of model updates. Only such evaluation can determine whether the architecture produces a genuine scientific benefit.

Priority research questions include validation of maturity indicators, relationships between interoperability and outcomes, methods for measuring context preservation, institutional answerability, low-resource deployment, software contingency, and the safety of AI-assisted evidence synthesis.

AI-EVIDENCE-02A adds specific questions: can AI improve extraction and contextualisation without increasing hidden bias; can claim-to-source lineage be maintained reliably; can patient applicability be represented without false precision; how should disagreement between AI and human reviewers be represented; and what validation is sufficient before clinical knowledge can influence care?

A medication-safety demonstrator should test the clinical knowledge architecture in a common primary-care situation such as hypertension plus type 2 diabetes with polypharmacy, including at least one fixed-dose combination. Evaluation should measure time to useful information, clinically important issues detected, false or

low-value alerts, missing monitoring, applicability explanations and clinician acceptance or rejection of suggestions.

A separate comparative medical-error study should examine how health systems detect adverse events, disclose and correct them, provide patients with explanation or redress, and convert incidents into safer care. Legal and compensation pathways should be studied as part of the ecosystem rather than used as a proxy for safety. No country-level conclusion is proposed in this edition; the immediate task is to define a reproducible protocol and gather comparable public evidence.

The study should also test whether digital maturity changes the burden of correction after an error. Relevant questions include whether records can be amended across connected systems, whether patients can discover which information was used, whether independent review is accessible, and whether lessons from one event are propagated into software, workflow, training or governance.

Key Messages

- IML now has both conceptual and implementation workstreams.
- Reference implementation is not equivalent to clinical validation.
- AI-EVIDENCE-02 should progress through explicit scaffold, test and validation stages.
- Scientific review and versioning are part of the architecture of trust.
- Future work should remain progressive, open and health-oriented.

ANNEXES

Annex A - Glossary of Core Concepts

Concept	Working definition
Health Information Ecosystem	The dynamic network of individuals, organisations, technologies, governance mechanisms and information flows that collectively generate, validate, exchange, interpret and use trustworthy health information to protect and improve health.
Continuity of Health Information	The capacity of health information to accompany the patient and the health system across time, settings and authorised actors without losing meaning, integrity or clinical relevance.
Knowledge Continuity	The capacity of successive clinical, public-health and research activities to build on previous validated knowledge rather than repeatedly restarting from fragmented information.
Clinical Context Preservation	The preservation of the clinical circumstances that give information its meaning, including symptoms, diagnosis, indication, treatment, evolution and relevant risk factors.
Institutional Engagement	The willingness and capacity of organisations to participate in dialogue, cooperation, governance and implementation activities supporting interoperability.
Institutional Readiness	The practical preparedness of an organisation or ecosystem to collaborate, respond, align incentives, allocate resources and participate in interoperable processes.
Institutional Answerability	The demonstrated capacity to receive, examine and substantively answer documented issues and to act on authoritative findings, warnings or recommendations.
Institutional Documentary Reliability	The capacity to maintain official documentary references that remain accessible, precise, current, attributable and correctable over time.
Payer Interoperability	The capacity of public insurers, national health services, private insurers and mutual organisations to exchange appropriate information with clinical and public-health actors under trustworthy governance.

Concept	Working definition
Open Clinical Workspace	A vendor-neutral, open-source-oriented clinical environment designed to interact with existing systems, preserve clinical context and provide progressively deployable reference services.
Information Stewardship	Collective responsibility for quality, security, traceability, proportionality, correction and usefulness of information throughout its lifecycle.
Progressive Interoperability	The principle that interoperability should improve step by step according to context, resources, priorities and measurable progress.
Database Quality	The capacity of database technologies and data stores to preserve trustworthy health information through integrity, security, traceability, reversibility, portability, correction, backup, restoration and maintainability.
Identity Access Layer	A secure trust and access layer that works with existing national, local or organisational identity systems to generate purpose-limited, auditable access mechanisms without creating a new personal identity number.
Contextual Access Token	A temporary, signed, auditable and revocable proof that identity or authorisation has been verified for a defined purpose, under defined rules and duration.
AI-EVIDENCE-01	The implemented IML scaffold for traceable scientific evidence retrieval and provenance across registered external evidence sources.
AI-EVIDENCE-02A	The proposed IML layer for patient-independent clinical evidence provenance, normalized claims, studied populations, certainty, applicability and medication-support knowledge while preserving source lineage and human validation.
Clinical Evidence Claim	A concise patient-independent clinical statement derived from one or more sources and linked to provenance, applicability, authority and review status rather than stored as an opaque publication summary.
Patient Applicability	The degree to which the population, conditions, intervention and context underlying a clinical claim correspond to the person and care situation in which the claim may be used.

Concept	Working definition
Application Priority	The order in which information should be considered for a clinical action because of jurisdiction, authority or binding status; it is distinct from scientific certainty.
Operational Medication Reference	A medication-information source designed for rapid practical use or software integration, such as a licensed national reference; operational usefulness does not replace regulatory authority.
Medication Decision Support	A governed process that combines active ingredients, patient context, contraindications, interactions, cumulative risk, dose constraints and monitoring requirements to support, not replace, prescribing judgement.
Medical Error Comparative Workstream	An IML research programme proposed to compare detection, disclosure, correction, redress and learning after healthcare harm across health systems without drawing premature country rankings or conclusions.

Annex B - Acronyms and Abbreviations

Acronym	Meaning
AI	Artificial Intelligence
AMR	Antimicrobial Resistance
BMR	Multidrug-resistant bacteria
DICOM	Digital Imaging and Communications in Medicine
EHR	Electronic Health Record
FHIR	Fast Healthcare Interoperability Resources
GDHM	Global Digital Health Monitor
GRADE	Grading of Recommendations Assessment, Development and Evaluation
HL7	Health Level Seven International
IHE	Integrating the Healthcare Enterprise
IML	Interoperability Maturity Lab
LOINC	Logical Observation Identifiers Names and Codes
MDR	Multidrug-resistant

Acronym	Meaning
NCT	ClinicalTrials.gov identifier
PMCID	PubMed Central Identifier
PMID	PubMed Identifier
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
SNOMED CT	Systematized Nomenclature of Medicine - Clinical Terms
UTI	Urinary Tract Infection
WHO	World Health Organization
EMA	European Medicines Agency
ePI	Electronic Product Information
PICO	Population, Intervention, Comparator and Outcomes
SmPC	Summary of Product Characteristics

Annex C - Proposed IML Indicator Catalogue

ID	Domain	Draft indicator
GOV-1	Governance and Standards	National/regional interoperability governance structure with defined responsibilities.
GOV-2	Governance and Standards	Implementation guides for priority information exchange processes.
GOV-3	Governance and Standards	Transparent processes for updating standards, terminology and governance rules.
TEC-1	Technical Interoperability	Secure APIs or exchange services for priority clinical information.
TEC-2	Technical Interoperability	Exchange of laboratory, prescription, discharge and imaging metadata using recognised structures.
TEC-3	Technical Interoperability	Testing environments, sandboxes or conformance mechanisms.

ID	Domain	Draft indicator
TEC-4	Technical Interoperability	Operating environments assessed for lifecycle support, portability and maintainability.
TEC-5	Technical Interoperability	Database technologies documented with data models, export, migration, backup/recovery and audit capabilities.
IDT-1	Identity, Consent and Trust	Reliable patient identification across participating organisations.
IDT-2	Identity, Consent and Trust	Operational consent or lawful-basis mechanisms appropriate to data use.
ADP-1	Adoption and Use	Routine use of interoperable workflows rather than pilot-only availability.
ADP-2	Adoption and Use	Training and support mechanisms for professionals and patients.
SEC-1	Security and Resilience	Defined cybersecurity and incident-response procedures for interoperable exchanges.
SEC-2	Security and Resilience	Auditability and traceability of access to sensitive information.
SEC-3	Security and Resilience	Patch management, hardening, audit and recovery for clinical/infrastructure operating systems.
SEC-4	Security and Resilience	Database access control, encryption where appropriate, audit logs, role separation, tested backup and recovery.
LRN-1	Feedback, Correction and Learning	Mechanisms for detecting and correcting erroneous information across systems.
LRN-2	Feedback, Correction and Learning	Use of interoperability, outcome, incident and feedback indicators for continuous improvement.

ID	Domain	Draft indicator
LRN-3	Feedback, Correction and Learning	Correction, versioning and traceable reuse of database-held information.
LRN-4	Feedback, Correction and Learning	Institutional documentary reliability and integrity over time.
LRN-5	Feedback, Correction and Learning	Institutional responsiveness and answerability to documented inquiries and authoritative findings.
INS-1	Institutional Engagement	Identified institutional contact points for scientific and operational collaboration.
INS-9	Institutional Engagement	Defined procedures and contact pathways for external proposals or inquiries, distinct from actual follow-through.
PAY-1	Payer Interoperability	Defined pathways for appropriate information exchange between clinical actors and payers.
AI-1	AI Readiness	Documented governance for AI-assisted use of health information under human responsibility.

Annex D - Evidence Grading Template

Level	Category	Description
Level A	Direct evidence	Official legal texts, standards, peer-reviewed studies, validated datasets or authoritative primary documents directly supporting the proposition.
Level B	Documented implementation	Public implementation reports, official dashboards, technical documentation or audited programme information.
Level C	Expert-informed evidence	Structured expert interviews, professional consensus or institutional responses, clearly identified as such.

Level	Category	Description
Level D	Provisional or indirect evidence	Indirect, incomplete or weakly attributable evidence requiring caution and further validation.

This source-quality template complements, but does not replace, the E0-E5 implementation-evidence ladder. The first describes source authority; the second describes maturity of demonstrated implementation.

Annex E - AMR/BMR Minimum Clinical Dataset

Field group	Elements
Patient/encounter context	Pseudonymous or authorised identifier, age/sex where relevant, encounter setting/date, relevant risk context.
Symptoms	Dysuria, frequency, urgency, suprapubic pain, flank pain, fever/systemic signs or explicit absence/unknown.
Urinalysis	Leucocyturia and other available relevant findings with method where necessary.
Culture	Specimen, collection date, organism, quantitative result/bacterial count, contamination indicators.
Susceptibility	Antimicrobial agents, interpretation, testing method/standard and date.
Clinical interpretation	Contamination, colonisation, asymptomatic bacteriuria, symptomatic UTI, complicated infection, uncertain/other.
Treatment	Drug, timing, dose/route where material, changes, reason where documented.
Outcome	Improvement, persistence, recurrence, complication, admission, follow-up unknown.
Provenance/correction	Source, timestamps, version, correction state, responsible actor.

Annex F - Payer Interoperability Working Indicators

- Ability to identify longitudinal care episodes across providers under lawful governance.
- Defined pathways for clinical-payer exchange without treating reimbursement coding as complete clinical truth.
- Mechanisms for prevention and chronic-care information when clinically relevant.
- Transparent correction and appeal pathways for erroneous administrative information that affects care.
- Portability and auditability across public, statutory, mutual and private financing arrangements.

Annex G - Institutional Engagement, Readiness and Answerability

Construct	Question	Do not confuse with
Institutional Engagement	Are collaboration pathways, contact points and governance participation available?	Actual substantive answer to a specific documented problem.
Institutional Readiness	Can the organisation allocate resources, align workflows and sustain implementation?	Formal policy alone.
LRN-5 Answerability	Does the institution substantively answer and act on relevant evidence or authoritative findings?	Automated acknowledgement or existence of a contact form.
LRN-4 Documentary reliability	Are official references stable, current, attributable and correctable?	General website quality or publicity.

Annex H - AI-EVIDENCE-01 Reference Architecture

Object	Current role
iml_ai.evidence_source	Registry of evidence sources, access mode, capabilities and licence note; credentials excluded.
iml_ai.evidence_query	De-identified bibliographic query metadata optionally linked to encounter or diagnostic assessment.
iml_ai.evidence_source_run	One auditable execution of one source, including success, no-result and failure states.
iml_ai.evidence_record	Canonical bibliographic/scientific metadata with publication, OA and retraction fields.
iml_ai.evidence_identifier	PMID, PMCID, DOI, NCT, Cochrane or other persistent identifier.
iml_ai.evidence_query_result	Traceable relationship between query, source run and canonical record.
iml_ai.evidence_access_profile	Deployment-level access capability without storing passwords, cookies or API secrets.

Registered source classes

Code	Source	Role	Current status
PUBMED	PubMed / NCBI	Biomedical bibliographic discovery and identifiers	Registered; connector disabled pending validation

Code	Source	Role	Current status
EUROPE_PMC	Europe PMC	Bibliographic and open-access/metadata discovery	Registered; connector disabled pending validation
CLINICALTRIALS_GOV	ClinicalTrials.gov	Trial registry and results-status metadata	Registered; connector disabled pending validation
CROSSREF	Crossref	DOI-centred metadata and deduplication support	Registered; connector disabled pending validation
COCHRANE_PUBLIC	Cochrane Library public metadata	Systematic-review discovery and lawful link-out	Registered; public metadata only; connector validation required

Annex I - AI-EVIDENCE-02A Clinical Evidence Provenance and Applicability Architecture

Status: IML proposal / migration draft. No claim of implemented or clinically validated AI-EVIDENCE-02A or Medication Decision Support service is made in this edition.

The architecture deliberately does not treat publications as the final operational knowledge object. It separates source discovery and access from clinical-claim extraction, provenance, population description, outcome-level certainty, patient applicability and consultation-time decision support.

Layer	Required traceability	Clinical-control principle
Source registry and policy	Source identity, jurisdiction, access/reuse basis, authority and retrieval route.	National binding/regulatory information can take priority for action without being labelled stronger science.
Clinical evidence claim	Normalized statement, type, dates, jurisdiction, review status and source links.	Claim is reviewable and not a hidden model conclusion.
Provenance	Identifiers, source locator/version, sponsor context, protocol/registration and extraction method.	Source can be revisited and conflicts remain visible.
Studied population	Inclusion/exclusion, age, comorbidity, renal/hepatic context, concomitant therapy and follow-up.	External validity is explicit.
Outcomes / GRADE	Outcome definition, effects and certainty plus methodological domains where available.	IML does not auto-author authoritative GRADE ratings.

Layer	Required traceability	Clinical-control principle
Applicability criteria	Factors that support, limit, require or exclude application of the claim.	Patient-specific match remains local and clinician-reviewable.
Medication product decomposition	Product-to-active-ingredient composition including fixed-dose combinations.	Interaction and duplication logic operates on active ingredients.
Medication decision rules	Contraindication, dose, interaction, cumulative-risk and monitoring rules with provenance.	Rules are patient-independent until runtime.
Consultation-time resolution	Patient context, triggered rules, rationale, source drill-down and clinician action.	AI proposes; clinician validates/rejects/modifies. Patient-specific cloud persistence is not implied.

External initiatives have defined roles rather than being absorbed into IML: GRADE supplies an established certainty methodology; MAGIC Evidence/MAGICapp can provide digitally structured PICO, evidence-profile and recommendation objects; EMA ePI/SmPC supplies regulatory product information; EMA Clinical Data Publication can provide deeper trial provenance. Country layers can add applicable regulatory, operational and decision-support sources such as VIDAL in France or FASS, Janusmed, Läkemedelsboken, Läkemedelsverket and Socialstyrelsen in Sweden, subject to licensing and technical validation.

Annex J - Technical Status Matrix

Component	Status in this edition	Interpretation
Clinical core	Active reference implementation work	Clinical structures exist; not a complete validated product.
PostgreSQL reference data foundation	Active reference implementation work	Used to test integrity, provenance, migration and deployment.
KSH97-P encounter reasons	Active module	One principal and one optional secondary reason per encounter in current module registry.
Terminology bridge	Active module	Connects clinical records to canonical terminology model.
Terminology translation workflow	Active module	Review queue; no automatic publication as official labels.
ICD-11 readiness	Scaffold	Prepares multi-classification/shadow coding; no live ICD-11 claim.
Local structured clinical extraction	Active module	Candidate fact extraction for clinician review.
Generic AI assistance	Scaffold	Suggestions remain explicit and human-validated.

Component	Status in this edition	Interpretation
AI-EVIDENCE-01	Scaffold v0.1.1	Traceable retrieval foundation; external connectors disabled until validation.
AI-EVIDENCE-02A	Proposed architecture / migration draft v0.3.2	Clinical evidence provenance, applicability and source-policy layer; not yet a validated clinical service.
France pack	Active workstream	HPRIM-first lab integration and national-service adaptation scope.
Guatemala pack	Reference adaptation workstream	Spanish/local-resource adaptation target.
Medication Decision Engine	Proposed scaffold	Active-ingredient decomposition and patient-independent medication rules are specified; patient-specific clinical use requires separate implementation and validation.

Annex K - Consolidated Change Log

Date / working state	Change	Scientific significance
Previous working draft	Health-first framework, six domains, GDHM complementarity, database/OS quality, identity-access layer, professional roles, Open Clinical Workspace, Responsible AI.	Established the 73-page conceptual base retained for this edition.
5 Sep 2026	Learning expansion to LRN-4 and LRN-5; France evidence and institutional answerability methodology.	Makes documentary reliability and actual follow-through explicit within Learning.
5 Sep 2026	Open Clinical Workspace implementation work reflected in project and repository.	Moves Part VIII from a purely proposed workspace toward a reference implementation workstream.
5 Sep 2026	AI-EVIDENCE-01 v0.1.1 retrieval/provenance scaffold and terminology canonicalisation.	Creates concrete traceable evidence and terminology foundations.

Date / working state	Change	Scientific significance
6 Sep 2026	AI-EVIDENCE-02 proposed architecture added to Founding Manuscript.	Extends provenance from retrieval into appraisal, synthesis, certainty inputs and clinical translation while retaining human validation.
6 Sep 2026	Full manuscript consolidation.	Restores a full-length Founding Manuscript rather than the earlier accidental condensed reconstruction.
6 Sep 2026	AI-EVIDENCE-02A redesigned around clinical evidence provenance, studied populations, applicability, GRADE/MAGIC/EMA integration and medication decision support.	Moves the unit of clinical knowledge from publication storage toward traceable, patient-independent claims that can be contextualised during care.
6 Sep 2026	Medical-error comparative research workstream added.	Frames medical error as a patient-safety, correction and learning problem and defines a comparative study of redress pathways without drawing conclusions at this stage.

Annex L - Suggested Citation and Review Template

Suggested citation:

Barrios R. Interoperability Maturity Lab (IML): A Scientific Framework for Trusted Health Information Ecosystems. Founding Manuscript. Full consolidated working draft, 6 September 2026. Open for Scientific Review.

Suggested review questions:

1. Does the manuscript clearly distinguish health objectives from technological means?
2. Is the Health Information Ecosystem concept sufficiently clear and operationally useful?
3. Are the six domains and cross-cutting dimensions coherent?
4. Do LRN-4 and LRN-5 improve the Learning domain without duplicating Institutional Engagement?
5. Are the country-profile evidence rules sufficiently transparent?
6. Is AMR/BMR appropriate as the first clinical demonstrator?
7. Does the Open Clinical Workspace remain vendor-neutral and realistically deployable?
8. Is the distinction between reference implementation and validated clinical service sufficiently explicit?
9. Does AI-EVIDENCE-01 preserve adequate provenance from query to retrieved record?
10. Is AI-EVIDENCE-02A appropriately separated from retrieval, regulatory authority and authoritative clinical recommendations?
11. Which AI-EVIDENCE-02A functions require formal human double review or independent validation?
12. Which claims require stronger literature support before external publication?
13. Does AI-EVIDENCE-02A adequately separate authority for action, operational retrieval, scientific certainty and applicability to the individual patient?
14. Is the proposed comparative medical-error workstream sufficiently neutral and reproducible to examine redress and learning without premature country conclusions?

Annex M - Proposed Comparative Study of Medical Error, Redress and Learning

Status: future research protocol / no comparative conclusion in this edition.

The proposed study starts from a patient-safety question rather than a legal ranking: after a person is harmed or may have been harmed by care, how effectively can the ecosystem detect the event, explain what happened, correct information, provide appropriate review or redress, and convert the event into safer future care? The study should use publicly available evidence first and distinguish formal legal availability from demonstrated practical accessibility.

Dimension	Comparative question
Detection and reporting	How can patients and professionals identify and report suspected harm, and what evidence exists that reports are investigated?
Disclosure and explanation	Are there defined processes for communicating what happened, uncertainty and subsequent corrections?
Record correction	Can erroneous clinical information be corrected and can corrections propagate across connected systems?
Independent review	What clinical, administrative, ombuds, mediation, expert or judicial review pathways are available and how accessible are they?
Compensation / redress	Which compensation or remedy mechanisms exist, what evidence describes eligibility, time, cost and practical burden, and which routes are non-judicial versus judicial?
Patient burden	What is known about the time, complexity, uncertainty and documentation burden imposed on the person seeking explanation or remedy?
Learning	Do incident findings lead to changes in practice, software, training, governance or regulation, and is follow-through traceable?
Evidence quality	Does the comparison rely on law/policy, operational data, audit, case outcomes or self-declaration, and how complete is the evidence?

IML should not assume that more litigation indicates lower safety, that fewer claims indicate higher safety, or that a statutory compensation mechanism proves effective access. Reporting cultures, legal traditions, transparency and data availability differ. The study should therefore describe systems before attempting any comparative interpretation.

The intended contribution is to test whether patient safety, correction, institutional answerability and information continuity can be analysed together. A later phase may examine whether differences in redress and learning mechanisms correlate with measurable improvements in quality of care; that hypothesis is not tested in the present manuscript.

Annex N - Selected References for Future Cited Versions

- World Health Organization. Global strategy on digital health 2020-2025. Geneva: WHO, 2021.
- Global Digital Health Monitor. Indicator Guide, methodology and country profile resources. Validation status should be rechecked at each IML edition.
- Organisation for Economic Co-operation and Development. Health Data Governance for the Digital Age. OECD Publishing, 2022.

- Agency for Healthcare Research and Quality. Learning Health Systems resources.
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- European Union. Regulation (EU) 2025/327 on the European Health Data Space.
- European Union. Regulation (EU) 2024/1183 establishing the European Digital Identity Framework.
- Page MJ et al. PRISMA 2020 statement and associated checklist/explanation resources.
- Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, eds. *Cochrane Handbook for Systematic Reviews of Interventions*, current online version.
- GRADE Working Group. GRADE Book / GRADE guidance, current online edition.
- NCBI. Entrez Programming Utilities (E-utilities) documentation.
- Europe PMC. RESTful Web Service documentation.
- ClinicalTrials.gov. API v2 documentation.
- Crossref. REST API documentation.
- Cochrane. Library and technical/public metadata documentation.

Editorial note. These references are targeted supports rather than an exhaustive literature review. Formal publication should separate sources supporting conceptual vocabulary, methodological indicators, technical architecture and empirical validation.

- World Health Organization. Global Patient Safety Action Plan 2021-2030. Geneva: WHO, 2021.
- GRADE Working Group. GRADE guidance and Evidence-to-Decision framework resources, current online edition.
- MAGIC Evidence Ecosystem / MAGICapp. Structured guideline, PICO, evidence-profile, GRADE and API resources, current online documentation.
- European Medicines Agency and European Medicines Regulatory Network. Electronic Product Information (ePI) Implementation Guide, version 1.0.0, 2026.
- European Medicines Agency. Clinical Data Publication / Policy 0070 resources, including clinical study reports, protocols and statistical-method documentation.
- VIDAL France. Medication reference, prescription-safety and software-integration documentation; access and reuse subject to applicable product terms.
- FASS. Swedish medication-information resources for healthcare professionals; product information and integration services.
- Region Stockholm / Karolinska University Hospital. Janusmed Interaktioner and Janusmed Riskprofil documentation.
- Swedish Medical Products Agency (Läkemedelsverket). Läkemedelsboken and national medicinal-product information resources.

Concluding Perspective

IML began from a simple observation: health information can be abundant and still fail to protect health when it is fragmented, decontextualised, poorly governed or difficult to correct.

The framework therefore places health before technology and treats interoperability as the capacity of an ecosystem to preserve trustworthy meaning across boundaries. Governance, standards, technology, identity, adoption, security and learning must be considered together.

The Open Clinical Workspace provides a practical test of these principles. Its value lies not in becoming another universal product, but in showing that a clinical core, terminology, connectors, country adaptations, audit and AI can be assembled transparently and progressively.

Artificial intelligence sharpens the same requirement. AI-EVIDENCE-01 preserves the retrieval trail. AI-EVIDENCE-02A proposes to preserve the path from source to clinically useful claim, studied population, certainty and patient applicability. Neither has value if provenance disappears at the moment an answer becomes fluent.

The same logic now extends to patient safety and medication decisions. IML should not confuse access to a publication with usable knowledge, nor a statistically precise population result with direct applicability to a particular person. The proposed evidence and medication layers aim to keep authority, provenance, uncertainty and patient context visible while reducing the time required to reach the information that matters in a consultation.

The long-term ambition remains modest in formulation but demanding in practice: information should become more trustworthy over time, institutions should be able to learn, and technology should remain an instrument placed at the service of people.